A trend analysis across five Drosophila EM connectomes
and five segmentation error models

BANC · FAFB · MANC · MCNS · MAOL

5 error models × 10 error rates × up to 5 replicate trials → 1,030 trials

Every number recomputed from the per-trial CSVs

20 August 2026

Contents

How this report is organised

This report analyses how five segmentation error models perturb the structure of five FlyWire connectomes as the error rate grows from 0 % to 20 %. Every figure and every number was recomputed from the per-trial result files.

§1 · Research Question

What this report answers and what you will find

Automated segmentation of electron-microscopy data is imperfect. Reconstruction errors are classified into five types: missed synapses (false negatives), false synapses (hallucinated connections), imprecise synapse-count measurement, split neurons (over-segmentation) and merged neurons (under-segmentation). Each corrupts the wiring diagram differently: some remove connections, some add them, some change only connection strengths, and some change the very identity of neurons.

This report asks: as the error rate grows from 0 % to 20 %, how strongly does each error type perturb the global structure of five FlyWire connectomes — and are those effects consistent across datasets?

Executive Summary

Topology-deleting errors (missed synapses, merge mis-segmentation) exert the most severe structural impact, scaling almost perfectly with the error rate. Measurement noise remains remarkably benign. Furthermore, all evaluated datasets exhibit qualitatively consistent responses, indicating that these structural responses are consistent across the evaluated connectomes.

Core Investigative Dimensions

- Direct: how do edge count, synapse count and degree respond as the error rate rises?
- Structural: are connectivity, component sizes and hub rankings preserved, or do they degrade?
- Comparative: which error type is most disruptive at the same nominal rate — and which is benign?

§2 · The Data

Five connectomes, five error models, 1,030 trials

The analysis covers five Drosophila EM connectomes: BANC (brain + nerve cord), FAFB (full adult female brain), MANC (male adult nerve cord), MAOL (male adult optic lobe) and MCNS (multi-cell-type nervous system). Each dataset was perturbed with up to five error models at ten error rates, with five independent seeded trials per rate, and every trial ran six graph analyses.

Table 1 · Baseline network properties at 0 % error (identical across all error models and trials)

Dataset	Neurons	Edges	Total synapses	Density	Reciprocity	Assortativity	Largest WCC	SCC count
BANC	158,262	3,990,039	23,556,214	0.00016	0.130	-0.046	153,952	18,345
FAFB	139,255	5,342,446	50,666,648	0.00028	0.165	-0.036	136,997	11,120
MANC	23,665	6,239,883	30,934,610	0.01114	0.285	+0.037	23,641	95
MCNS	166,694	6,239,112	89,797,169	0.00022	0.157	-0.057	165,820	8,859
MAOL	52,445	6,736,968	26,492,194	0.00245	0.278	-0.054	51,503	1,705

§2.2 · Data Coverage and Rate Grids

Which error models were run on which datasets

Table 2 · Replicate trials per rate per error model (— = not run)

Dataset	Missed syn.	False syn.	Syn. count	Split	Merge
BANC	5	5	5	5	5
FAFB	5	5	5	5	2
MANC	5	—	5	5	1
MCNS	5	5	5	5	5
MAOL	5	—	5	5	—

Three error-model runs are missing or partial: MANC and MAOL have no false-synapse run, MAOL has no merge-error run, FAFB merge has 2 trials/rate and MANC merge has 1 trial/rate. All other cells have the full 5 trials.

Two rate grids are used: missed/false synapses at 0, 0.25, 0.5, 0.75, 1, 2, 5, 10, 15, 20 %; synapse-count, split and merge errors at 0, 0.5, 1, 2, 3, 5, 7.5, 10, 15, 20 %. All values are means over the replicate trials.

What the six analyses measure

- basic_structure — node count, edge count, total synapses, weight mean/variance/std/max, density.
- degree_distribution — in/out/total degree mean, std and max.
- pagerank — weighted PageRank (damping 0.85); Pearson/Spearman correlation and top-100 overlap of perturbed vs baseline scores.
- assortativity — degree assortativity of the directed graph.
- connected_components — weak and strong component counts and largest sizes.
- reciprocity — fraction of edges with a reciprocal counterpart.

§3 · Method

How each error model perturbs the connectome graph

Table 3 · The five error models and what they do

Error model	Perturbation applied	Expected structural effect
Missed synapses	Removes synapses per edge with calibrated probability; an edge is deleted only when its weight reaches 0.	Edges ↓, weights ↓, degree ↓
False synapses	Adds k = round(rate × edges) new edges sampled from region-restricted, Jaccard-ranked candidate pairs, capped by candidate availability; weights from the weak-edge distribution.	Edges ↑, density ↑
Synapse count measurement	Adds Gaussian noise to edge weights, σ = rate·w, clamped ≥ 1; topology untouched.	Weights only; structure unchanged
Split errors	Splits round(rate × eligible) neurons (total degree ≥ 10) into two fragments; every edge is rewired exactly once.	Neurons ↑, edges preserved, degree ↓
Merge errors	Merges k = round(0.5·rate·eligible) region/soma-compatible neuron pairs ranked by shared partners; edges re-attached, parallels summed, A↔B dropped.	Neurons ↓, edges collapse, synapses ≈ conserved

What the nominal rate r means: the models are not directly comparable at the same r, because r is a different biological quantity in each model. Missed synapses remove a fraction r of synapses (removal probability is vulnerability-weighted, so per-edge removal is not uniform, but aggregate synapse loss matches r exactly). False synapses add k = round(r × edge count) new edges. Synapse-count measurement applies proportional noise σ = r·w to every weight. Split errors split a fraction r of eligible neurons (total degree ≥ 10). Merge errors absorb a fraction r of candidate-pool neurons into k = round(0.5 × r × eligible) pairs. All stochastic operations use recorded seeds, so every trial is exactly reproducible.

§3.2 · Effect Measure and Scale Conventions

How to read every trend figure in this report

How to read the figures

Every trend figure in §5 uses a unified layout for cross-dataset comparison: error rate (%) is plotted on the x-axis, and the relative percentage change of a metric against its 0 % baseline is on the y-axis. The formula is $\Delta\% = (m(\text{rate}) - m(0)) / |m(0)| \times 100\ \%$. A value of −20 % indicates a 20 % reduction from the baseline value. The dashed horizontal line marks zero change (no effect).

Scale conventions

- All evaluated connectome datasets are overlaid on a single axis for direct comparison.
- Direct endpoint labeling replaces traditional legends; dataset names and terminal values are positioned adjacent to each line.
- Automatic displacement prevents label occlusion when final values converge.
- PageRank is plotted as Pearson correlation (not % change) because the baseline is exactly 1.0. Coloured bands show quality thresholds.
- For split/merge errors the node set changes; PageRank vectors are re-aligned to baseline node space before correlating.

§3.3 · Statistical Caveats

Reading the trend figures and handling artefacts

Figure conventions

Figures in §5 use error rate (%) on the x-axis and relative % change Δ% against the 0 % baseline on the y-axis, calculated as $\Delta\% = (m(\text{rate}) - m(0)) / |m(0)| \times 100\ \%$. All datasets are plotted on unified axes to facilitate direct comparative analysis.

Statistical caveats

- Baseline Inflation: Assortativity has a near-zero baseline, so % change is inflated. We report it but exclude it from the heatmap.
- Metric Scaling: metrics whose absolute change stays below 0.25 % across all rates are treated as negligible. This is a descriptive magnitude threshold, not statistical significance.
- PageRank: Plotted as Pearson correlation (r) rather than % change. Pearson measures vector correlation, not rank preservation; Spearman correlation and top-100 overlap are reported in §5.5b.
- Split/Merge: PageRank vectors are re-aligned to baseline node space before correlating.
- Partial Runs: Cross-dataset means for merge errors include FAFB (2 trials) and MANC (1 trial).
- Uncertainty: trial-to-trial SD of the reported % changes stays below 0.18 pp in every 5-trial cell for the structural headline metrics (edge count, total synapses, mean degree, largest WCC/SCC, reciprocity); weight variance reaches 1.3 pp and assortativity 14.9 pp (near-zero baseline inflates relative spread). The MANC merge row (n = 1) has no trial variance.

§4 · Verification

Consistency checks performed before analysis

Table 4 · Data-quality verification

Check	What was tested	Outcome
Completeness	All six analyses present and successful in every trial	6,180 analysis rows; 0 failures
Baseline invariance	0 % runs identical across trials and error models	max relative difference 0.0 (bitwise identical)
Degree identity	sum(in-degrees) = edge count; in-degree mean = edges/nodes	max relative error 2.2e-16
Density identity	density = edges / (n·(n−1))	max relative error 7.0e-13
Missed synapses	removed synapses ≈ rate · synapses (QC enforced)	removed count matches target
False synapses	edge additions = min(planned k = round(rate · edges), candidates available)	exact for BANC/FAFB at all rates; MCNS caps at the candidate-pool size (18.2 % instead of 20 % at the top rate)
Syn. count noise	weight-variance increase ≈ rate²·E[w²] (theoretical prediction)	predicted 5.40 vs observed 5.45 (means across datasets)
Split/merge	edge count preserved (split) / collapse bookkeeping (merge)	matches plan metadata
PageRank alignment	split/merge vectors re-aligned to baseline node space	correlations in [0.5, 1] at all rates

The two structural identities (in-degree mean = edges/nodes and density = edges/(n·(n−1))) hold to machine precision, and the 0 % baseline is bitwise identical across trials and error models. All checks passed; the changes reported in §5 therefore reflect the perturbations themselves, not run-to-run variability.

§4.2 · Anomalies Investigated and Resolved

Suspicious patterns that were examined and explained

- 'False-synapse degree shift never varies across trials' — a mathematical identity, not a bug: false synapses add a deterministic number of edges every trial, so their trial-to-trial spread is exactly zero. Missed synapses remove a binomial-random number, so their spread is non-zero.
- 'Synapse-count measurement changes almost nothing' — by design it perturbs weights only; the variance increase matches theory ($rate^2 \cdot E[w^2]$), and structural metrics are completely untouched.
- 'Largest weak component is flat for false synapses' — new edges land inside the already-giant component, so adding them never enlarges it; the strong component does grow.
- 'Edge count is flat for split errors' — splitting rewires edges (each assigned exactly once) rather than creating them; the neuron count grows instead.
- No formula errors were found in the outputs. The only real caveat is run coverage: FAFB and MANC merge-error results are partial (2 and 1 trials per rate), and MANC/MAOL have no false-synapse run.

Conclusion: The datasets exhibit internal consistency and are structurally robust.

§5 · Results

Principal observations across all error models

Figures in this section show, for each combination of error model and metric, how that metric changes as a function of error rate. Each figure occupies one full landscape page for maximum readability: datasets are overlaid on a single axis with direct endpoint labeling to enable immediate cross-dataset comparisons. Lines approaching the zero axis indicate high structural resilience.

Principal observations (see Table 5)

- Missed synapses: synapses fall one-for-one with the rate (−20 % at 20 %); edges only −5 %; PageRank Pearson r 0.999.
- False synapses: edges and mean degree rise in step with the rate (+19 % at 20 %); PageRank Pearson r 0.994.
- Synapse-count noise: only the weight variance moves (+5.5 %); all topological metrics are untouched.
- Split errors: edges preserved exactly, but spread over more neurons → mean degree −15 %, largest weak component +18 %.
- Merge errors: most invasive to connectivity — edges −11 %, weight variance +47 %, largest weak component −9 %.

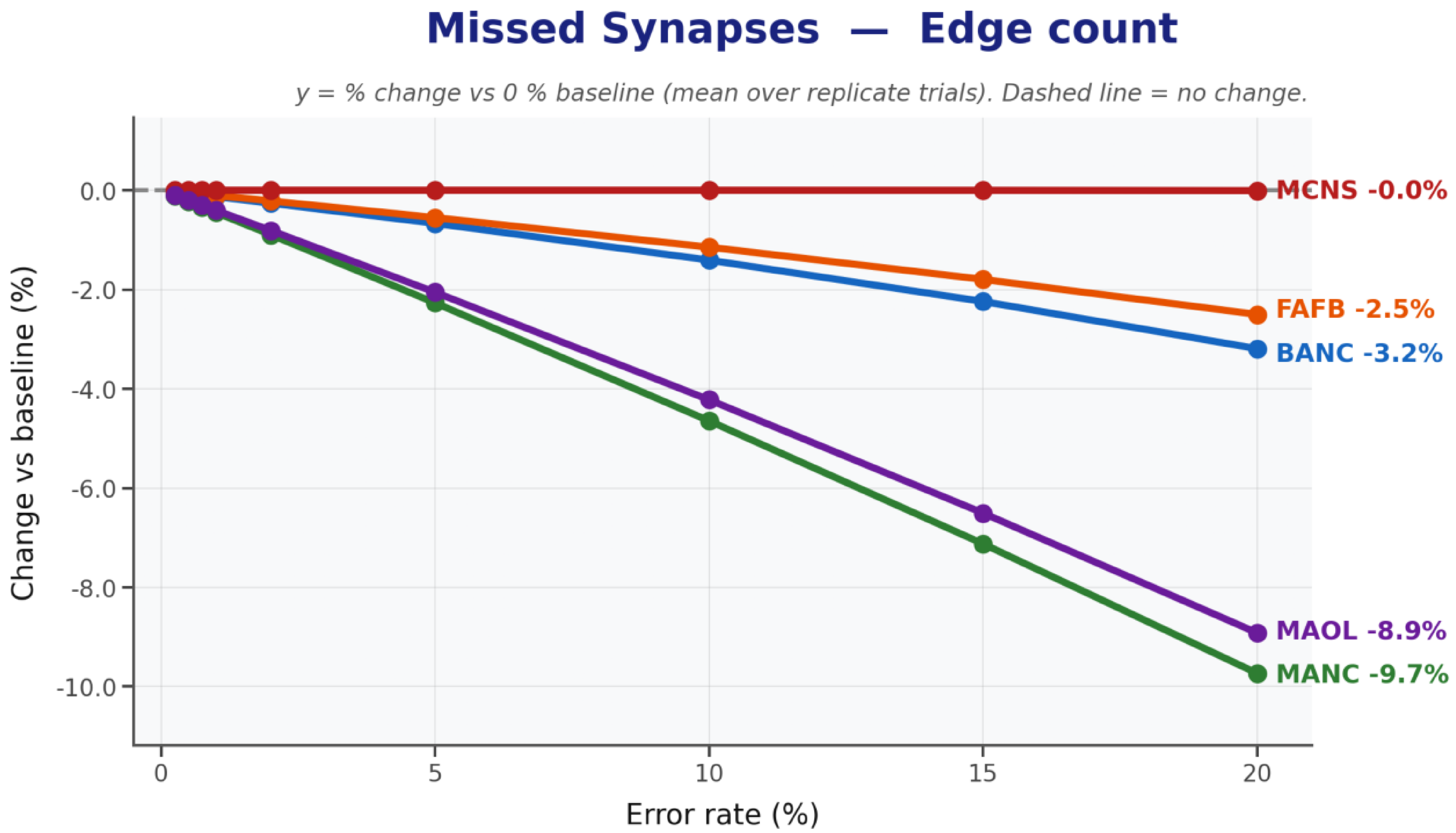


Figure 1 · Missed synapses — Edge count. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

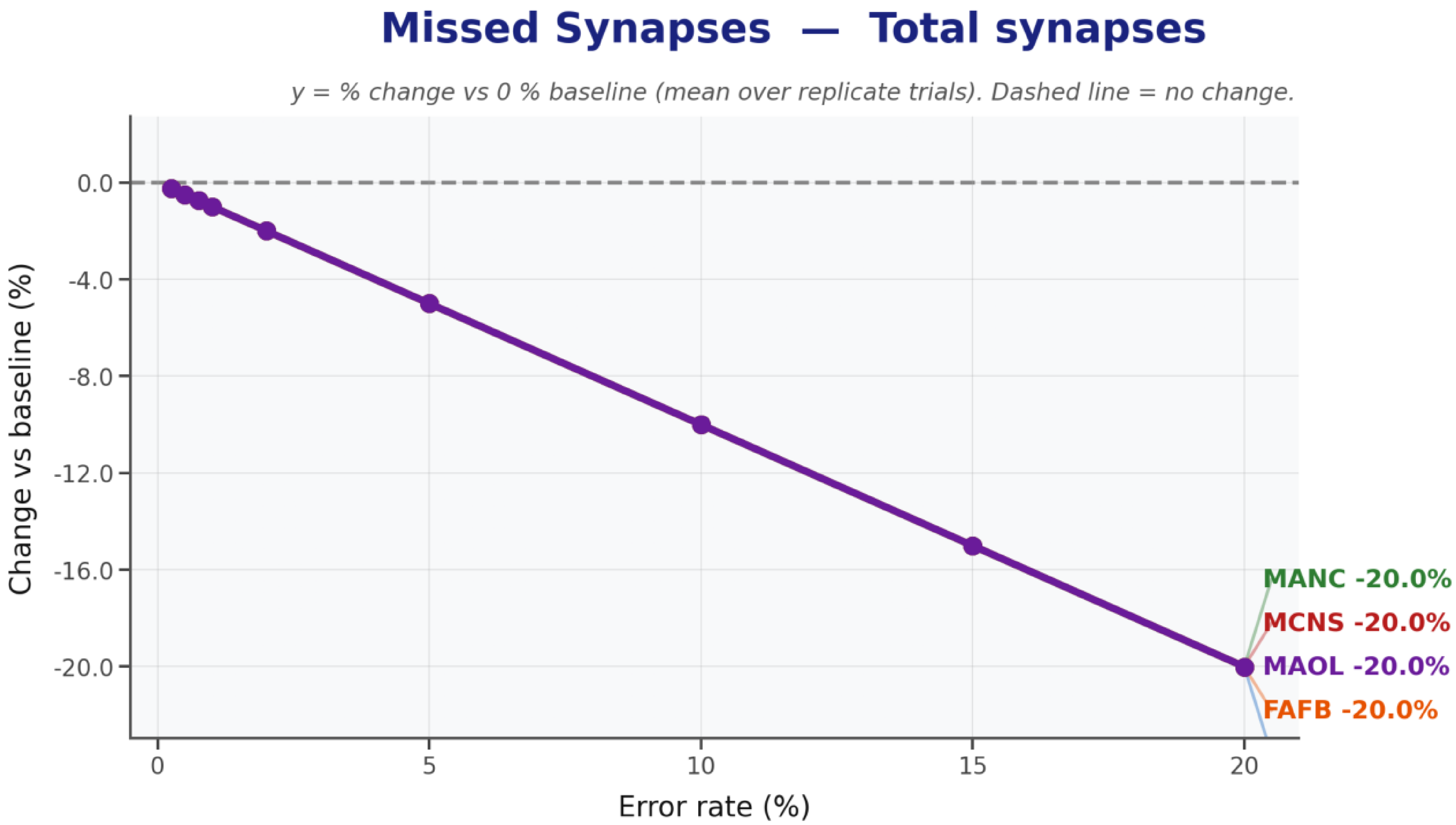


Figure 2 · Missed synapses — Total synapses. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

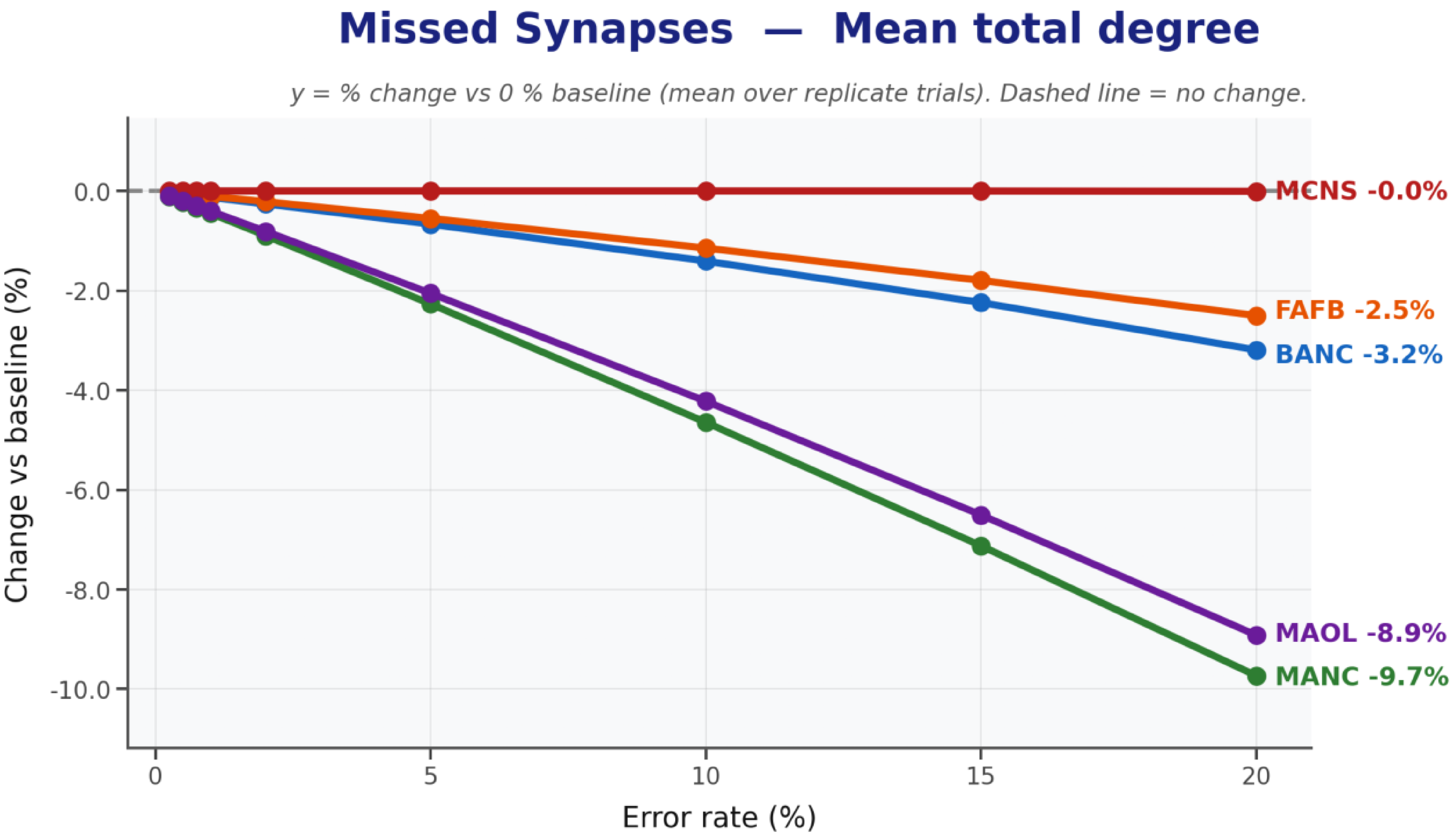


Figure 3 · Missed synapses — Mean total degree. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

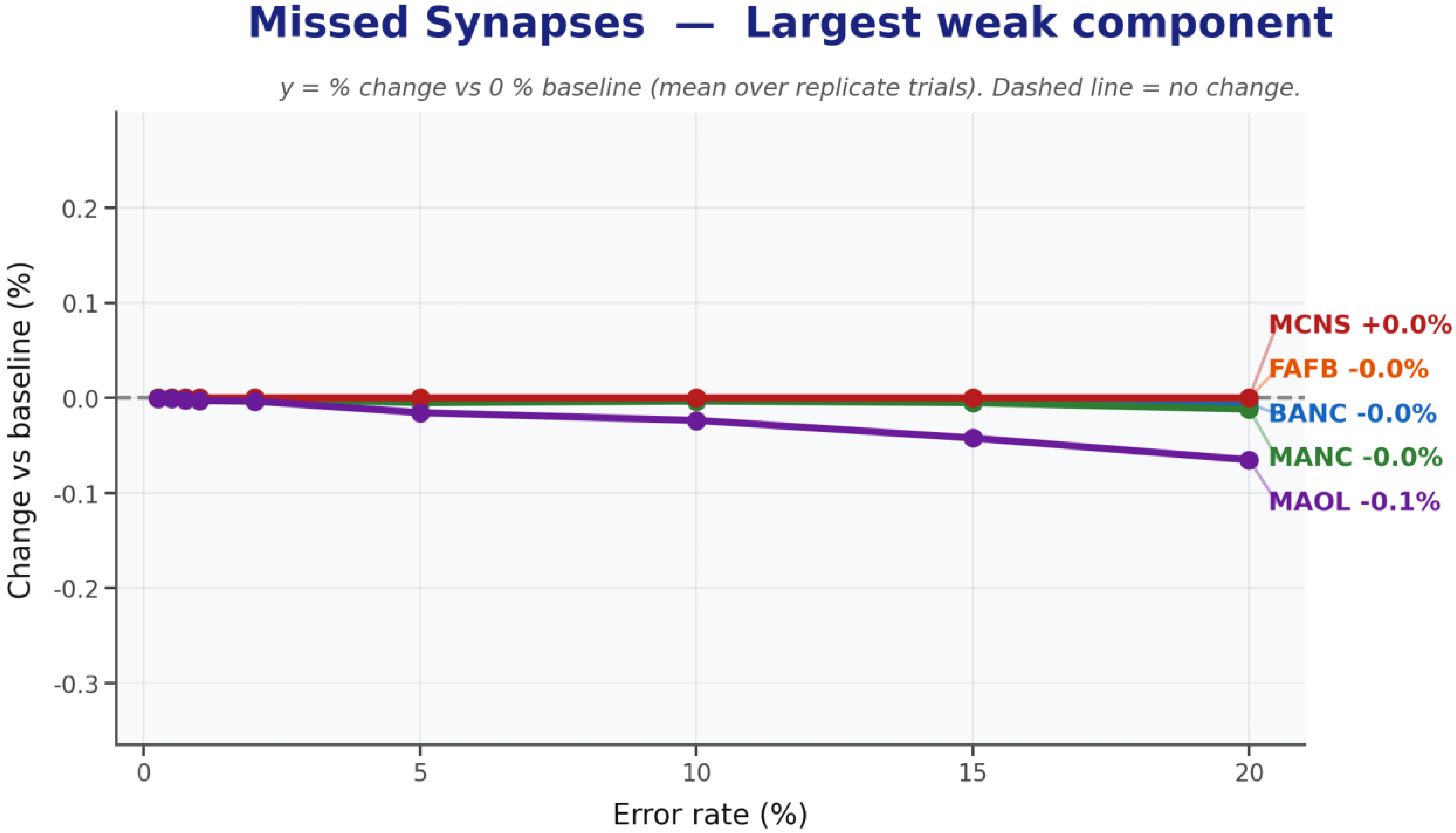


Figure 4 · Missed synapses — Largest weak component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

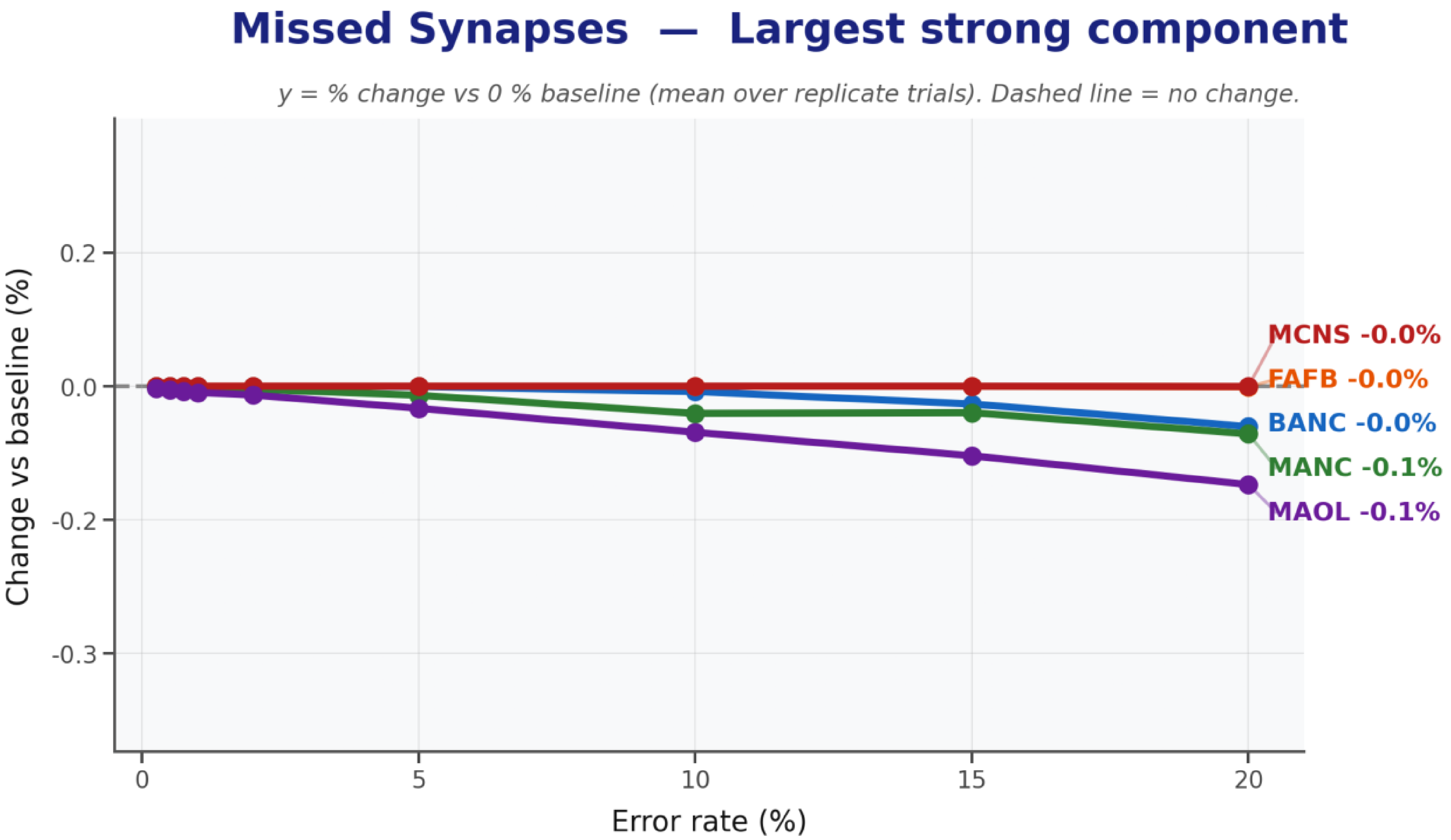


Figure 5 · Missed synapses — Largest strong component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

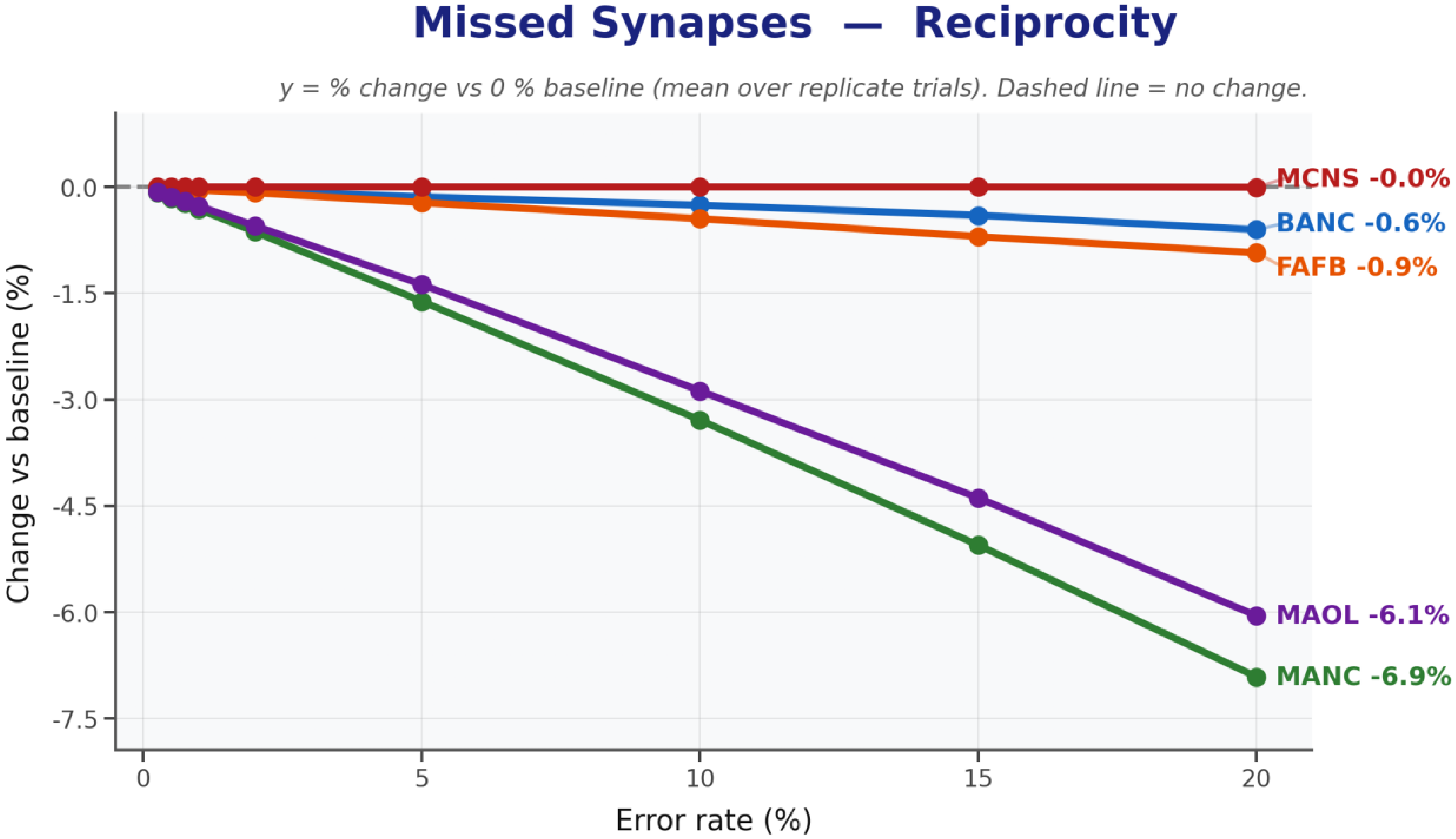


Figure 6 · Missed synapses — Reciprocity. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

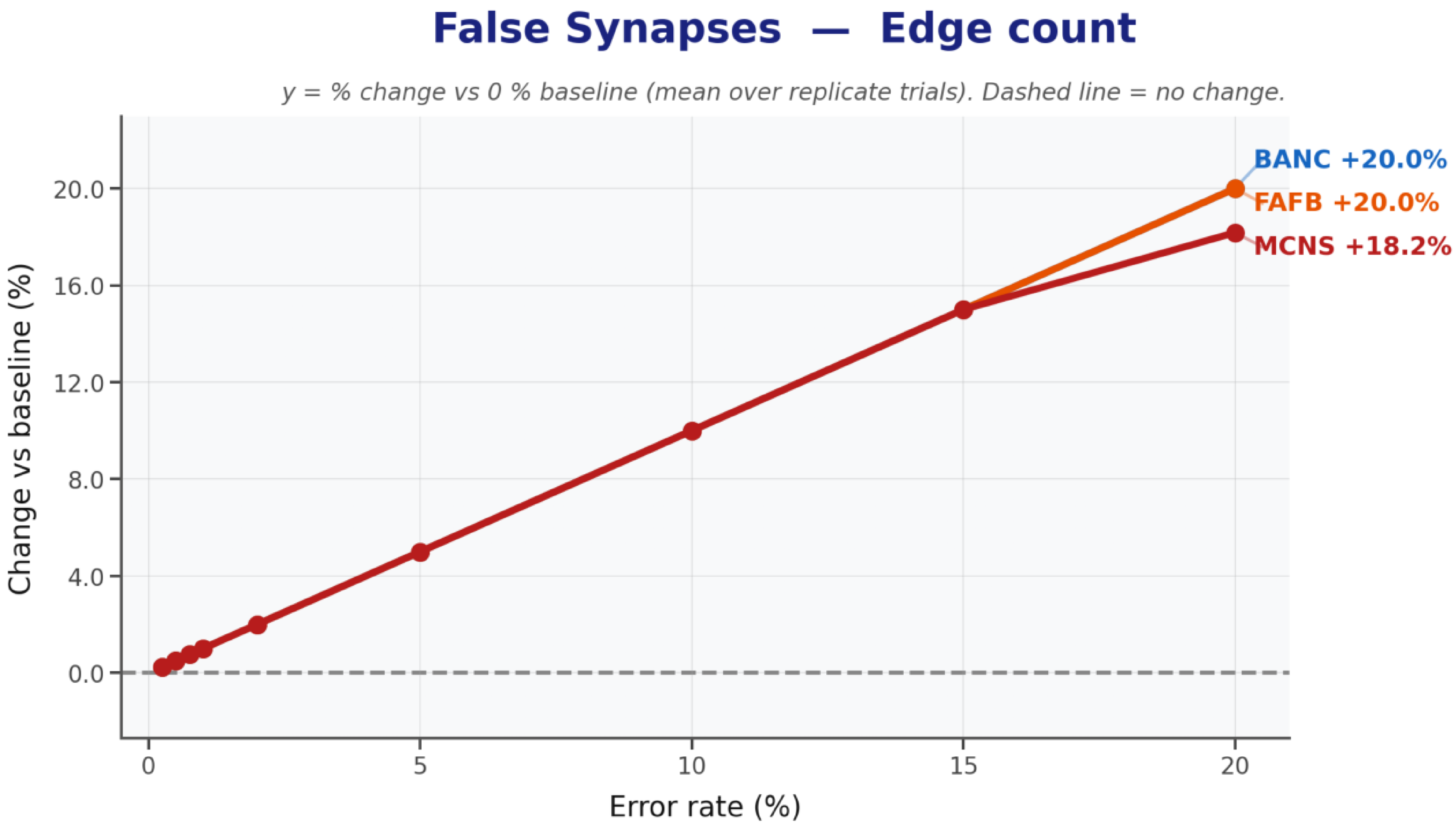


Figure 7 · False synapses — Edge count. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

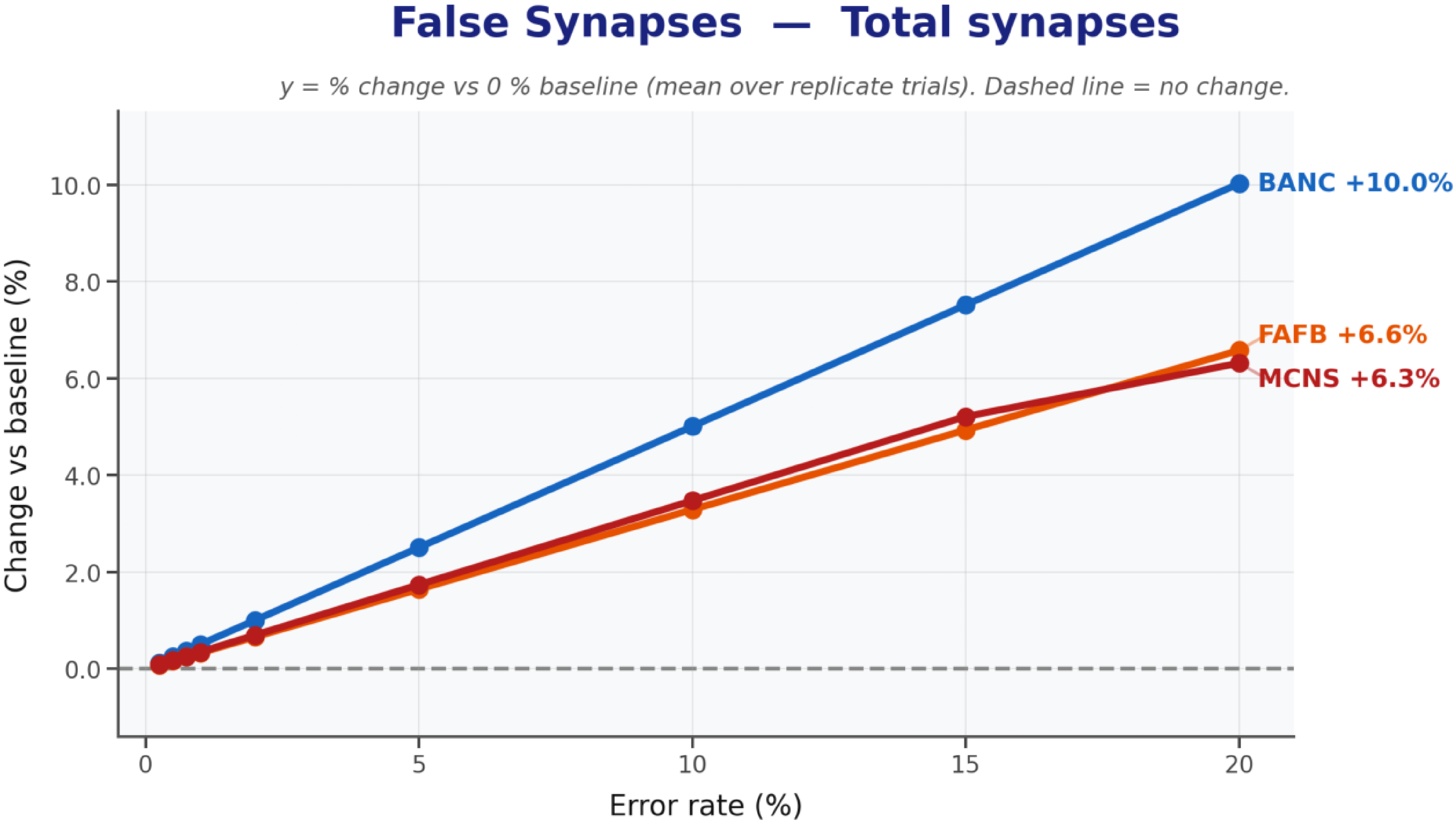


Figure 8 · False synapses — Total synapses. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

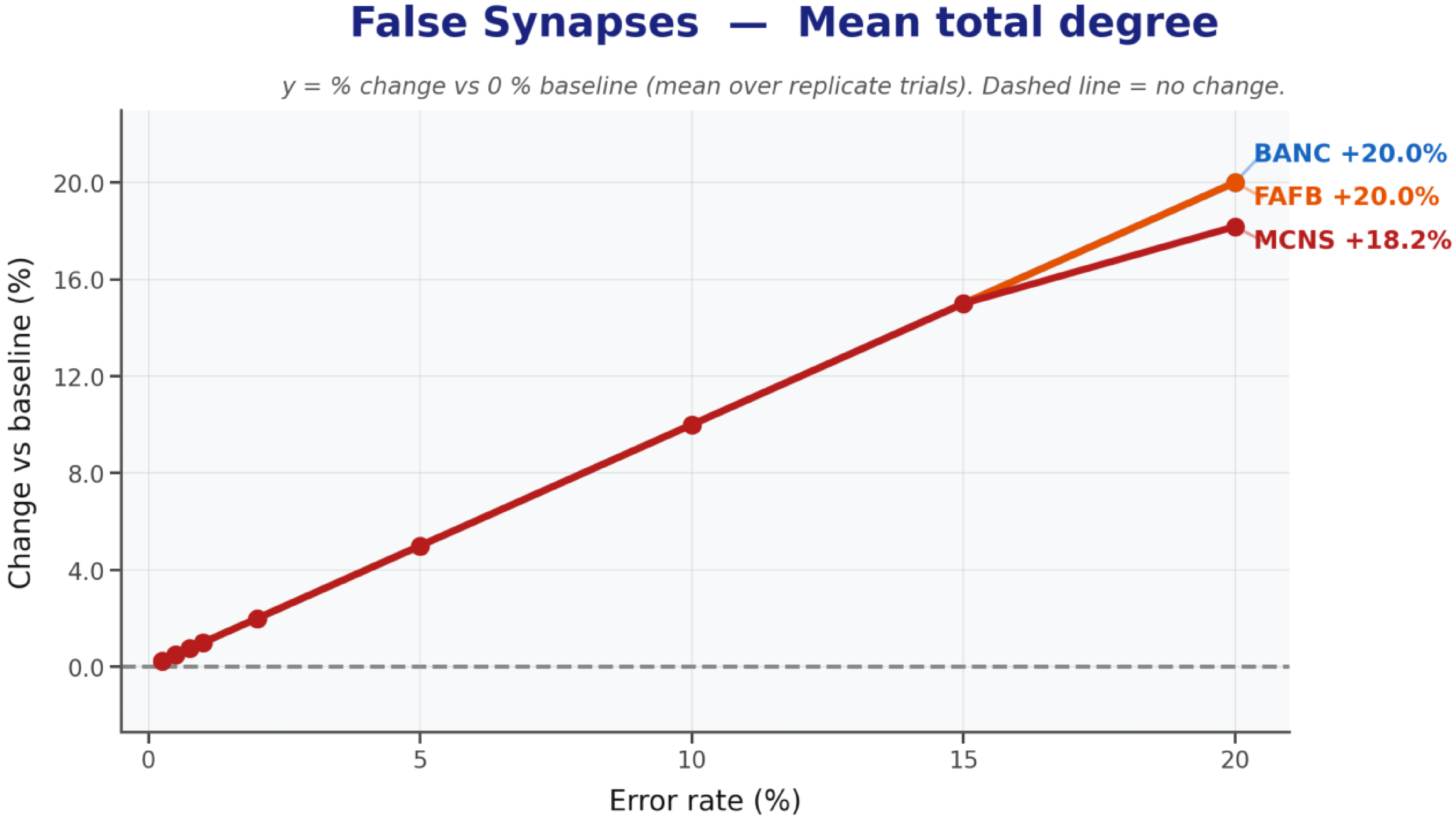


Figure 9 · False synapses — Mean total degree. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

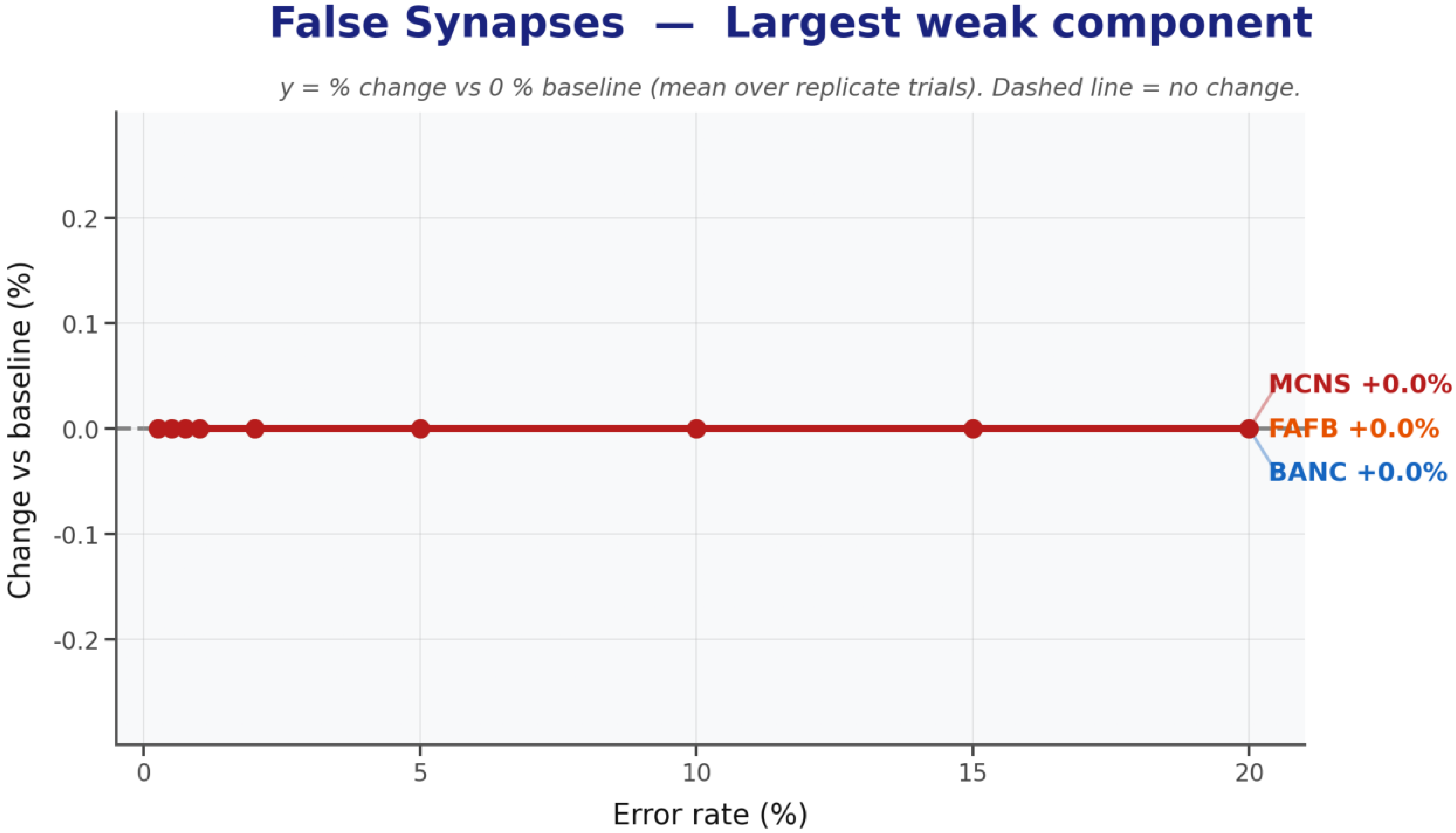


Figure 10 · False synapses — Largest weak component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

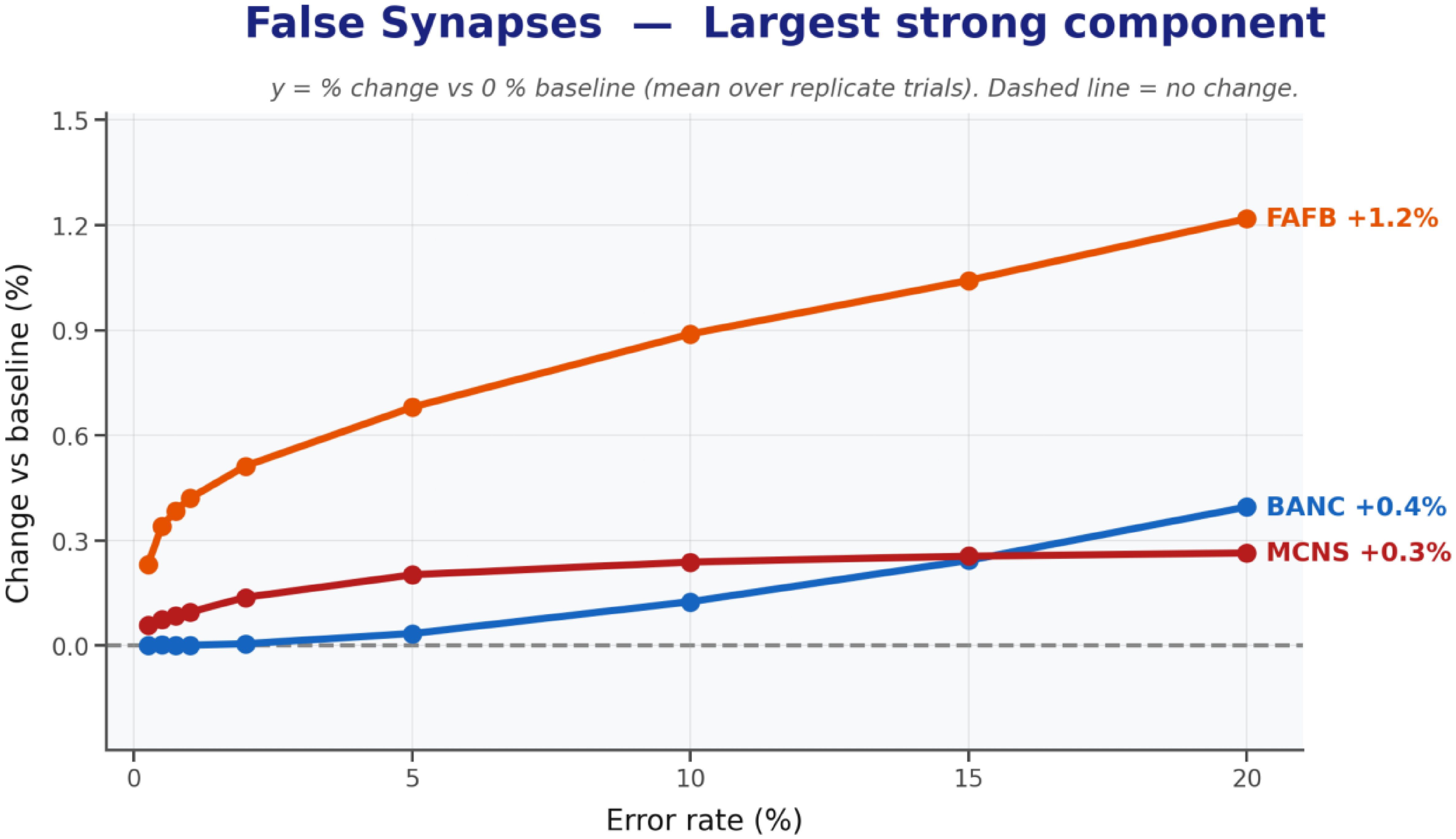


Figure 11 · False synapses — Largest strong component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

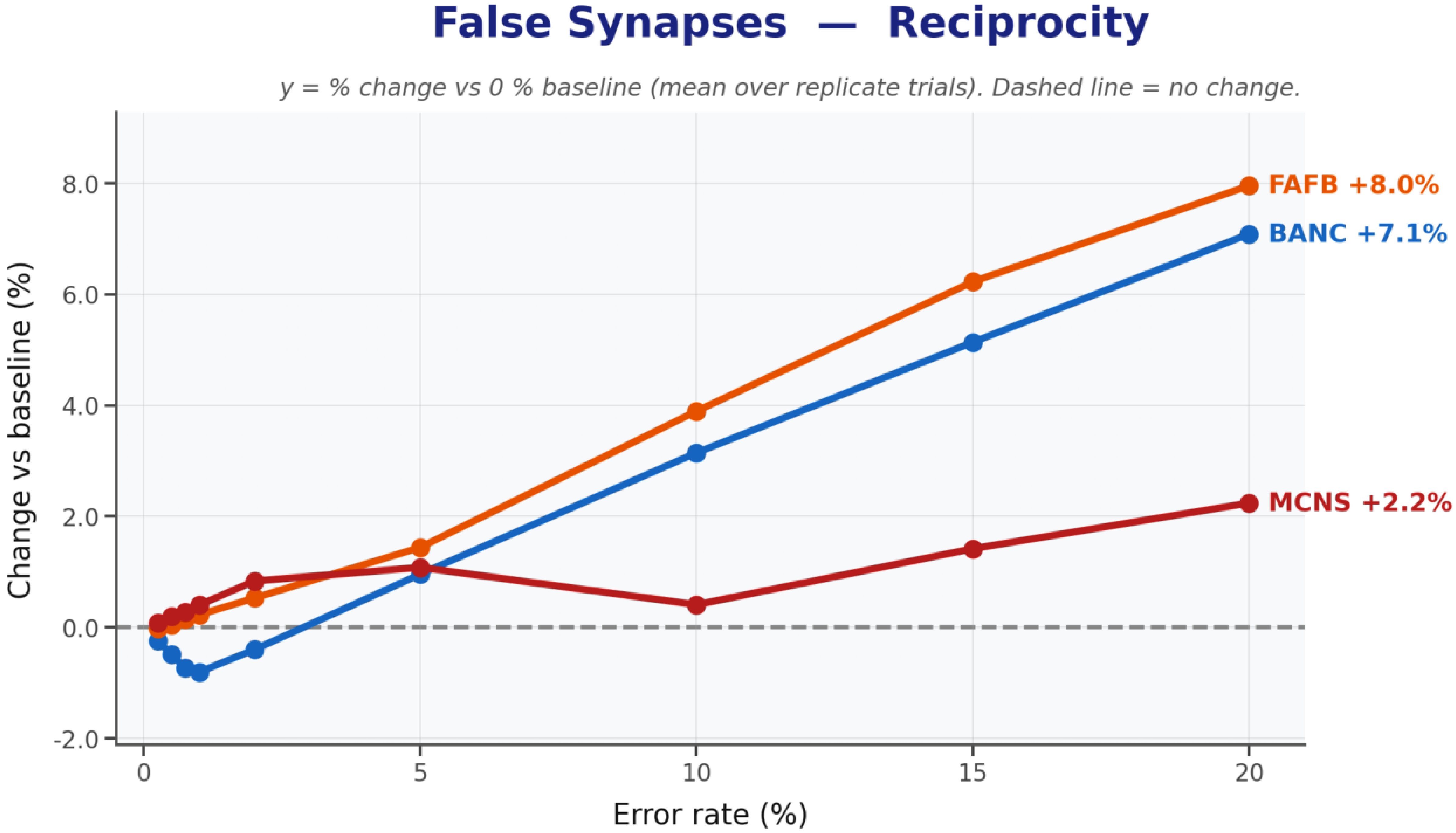


Figure 12 · False synapses — Reciprocity. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

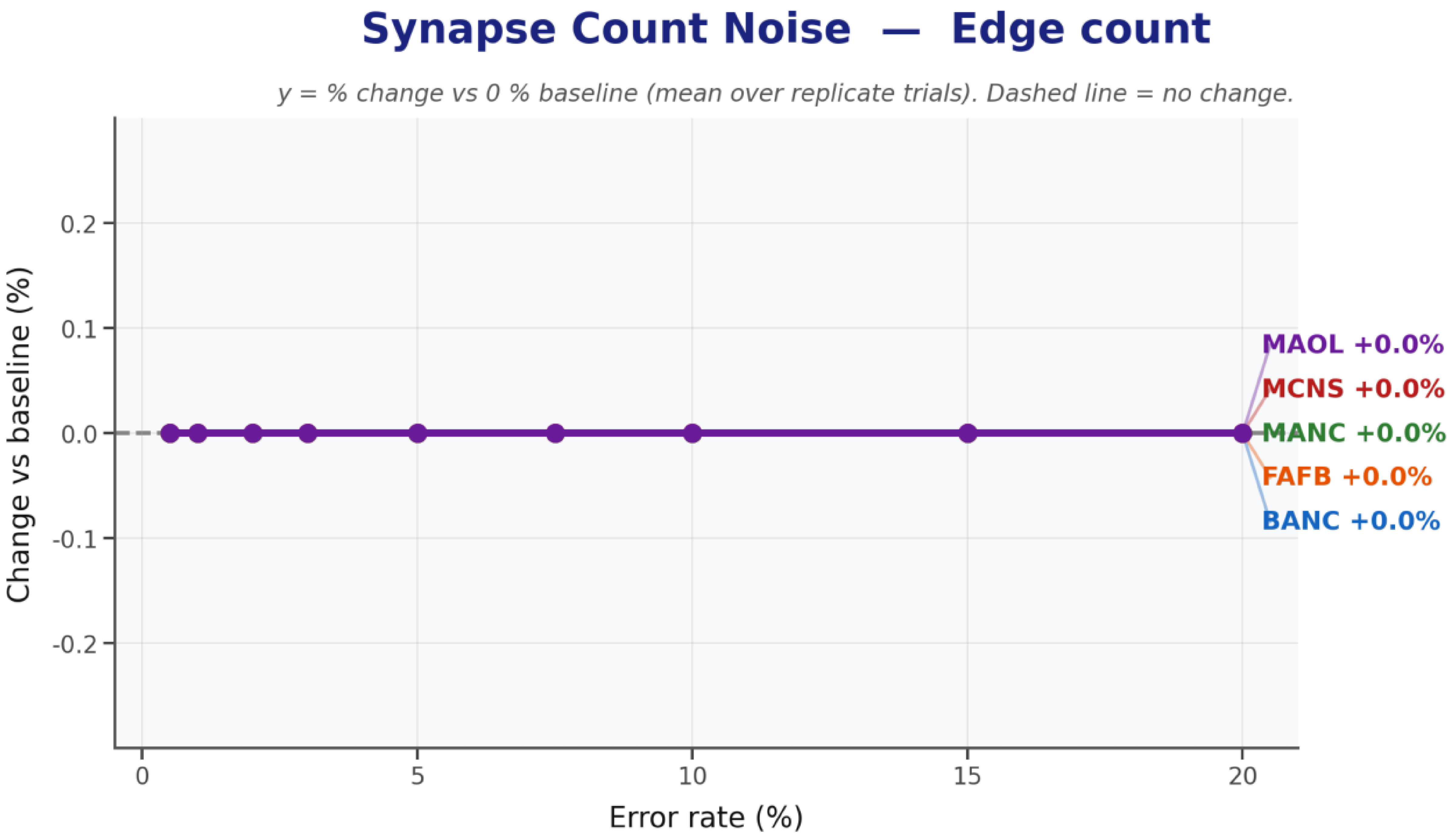


Figure 13 · Synapse count measurement — Edge count. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

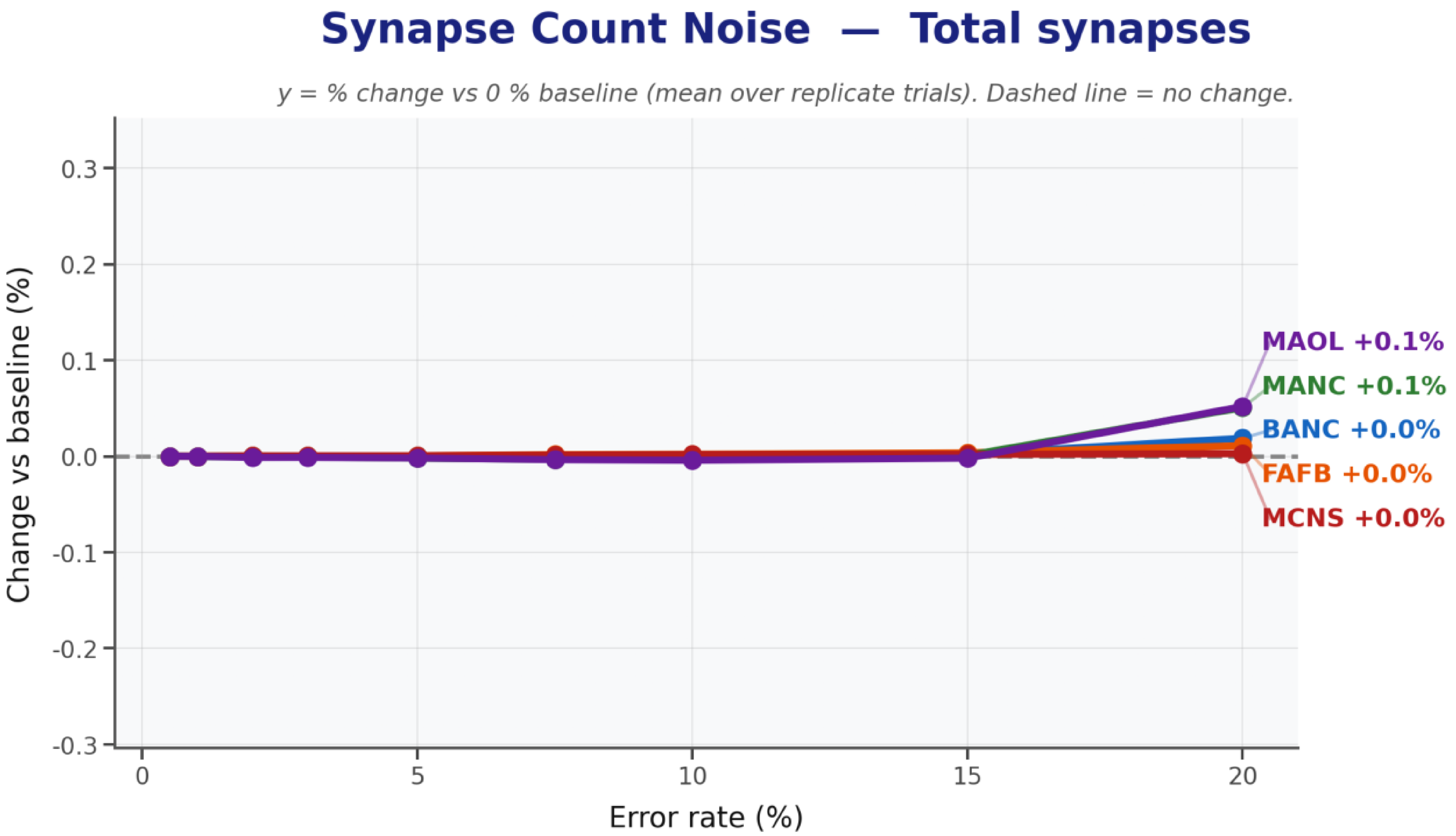


Figure 14 · Synapse count measurement — Total synapses. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

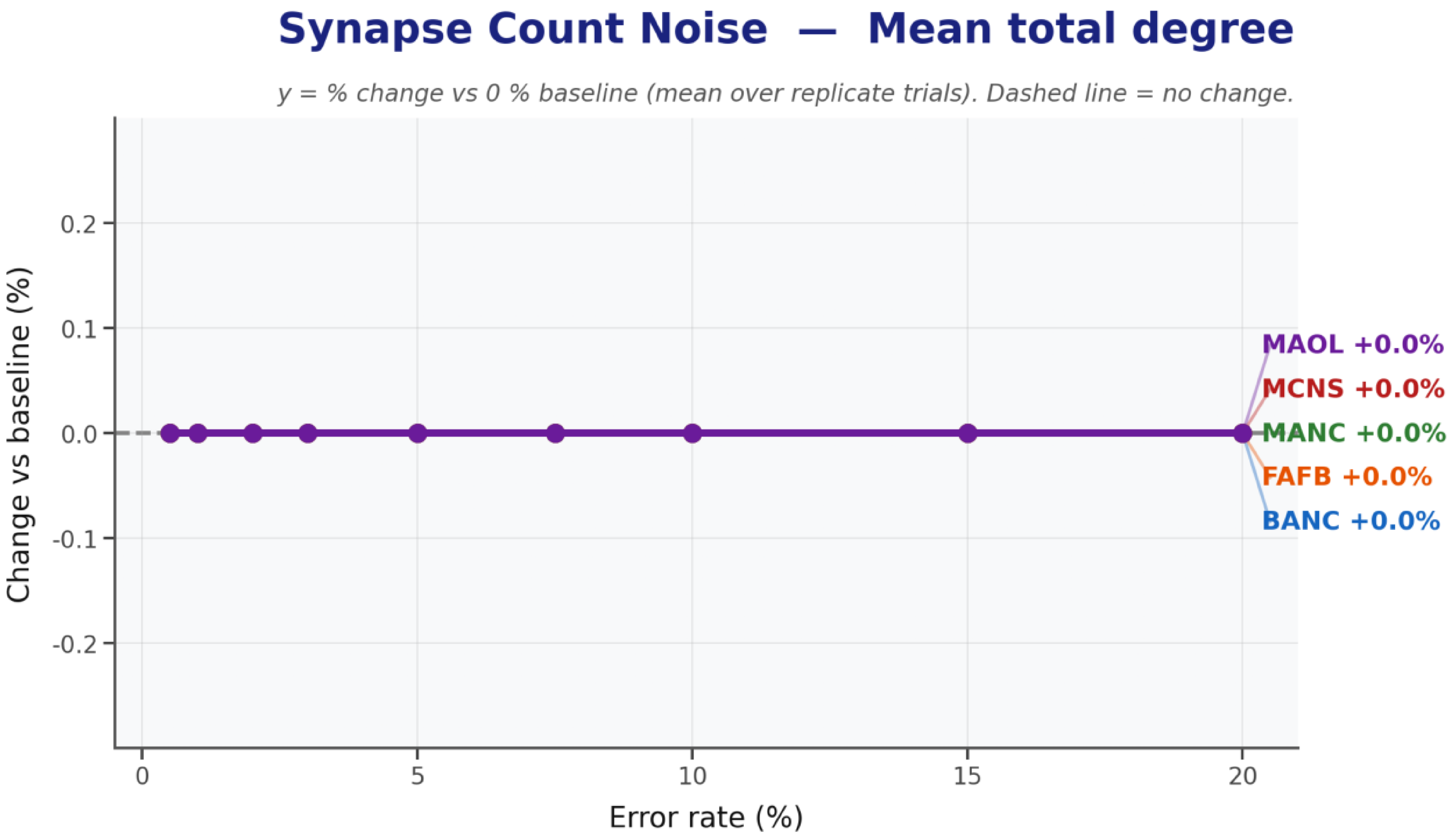


Figure 15 · Synapse count measurement — Mean total degree. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Synapse Count Noise — Largest weak component

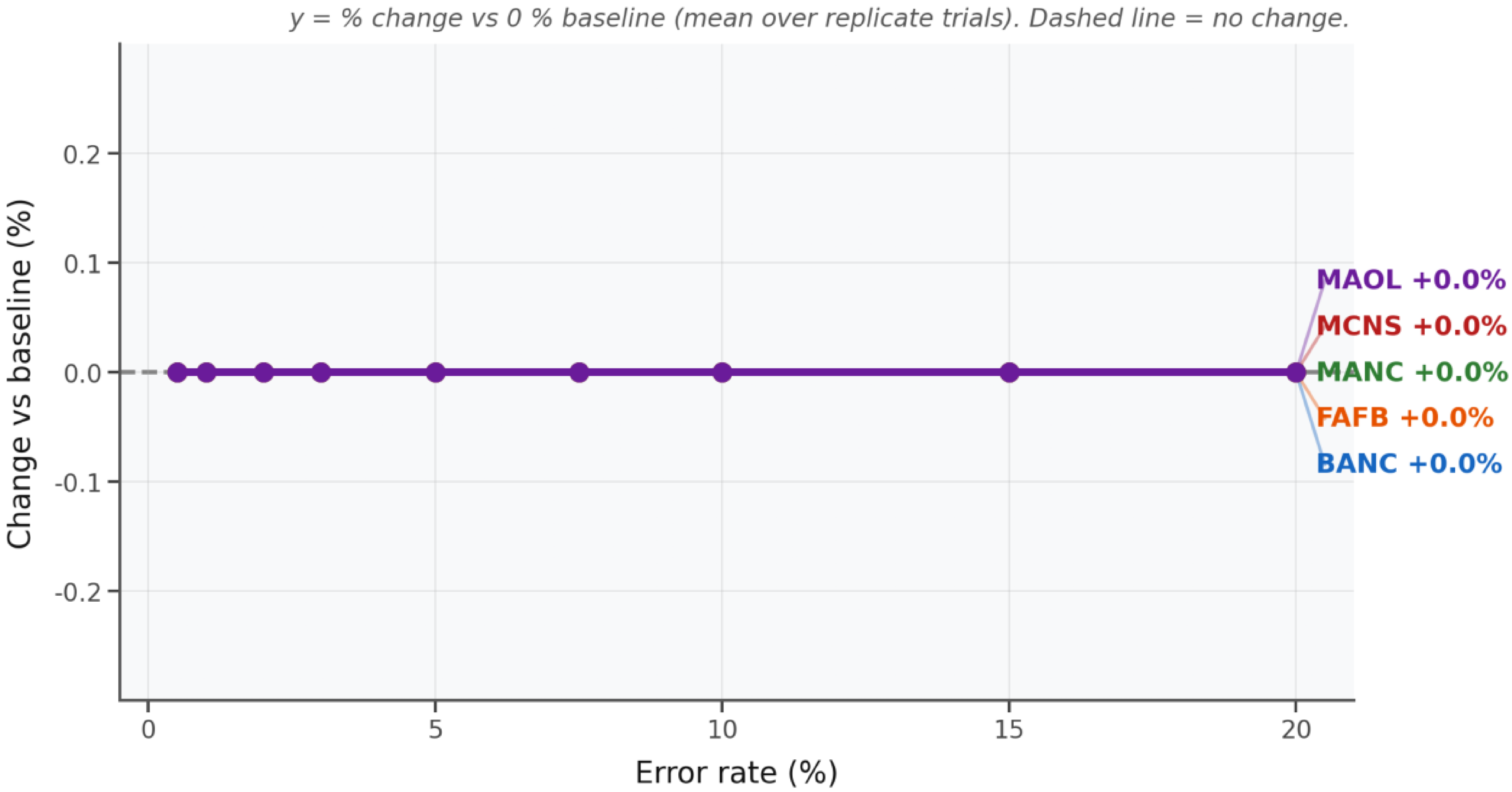


Figure 16 · Synapse count measurement — Largest weak component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

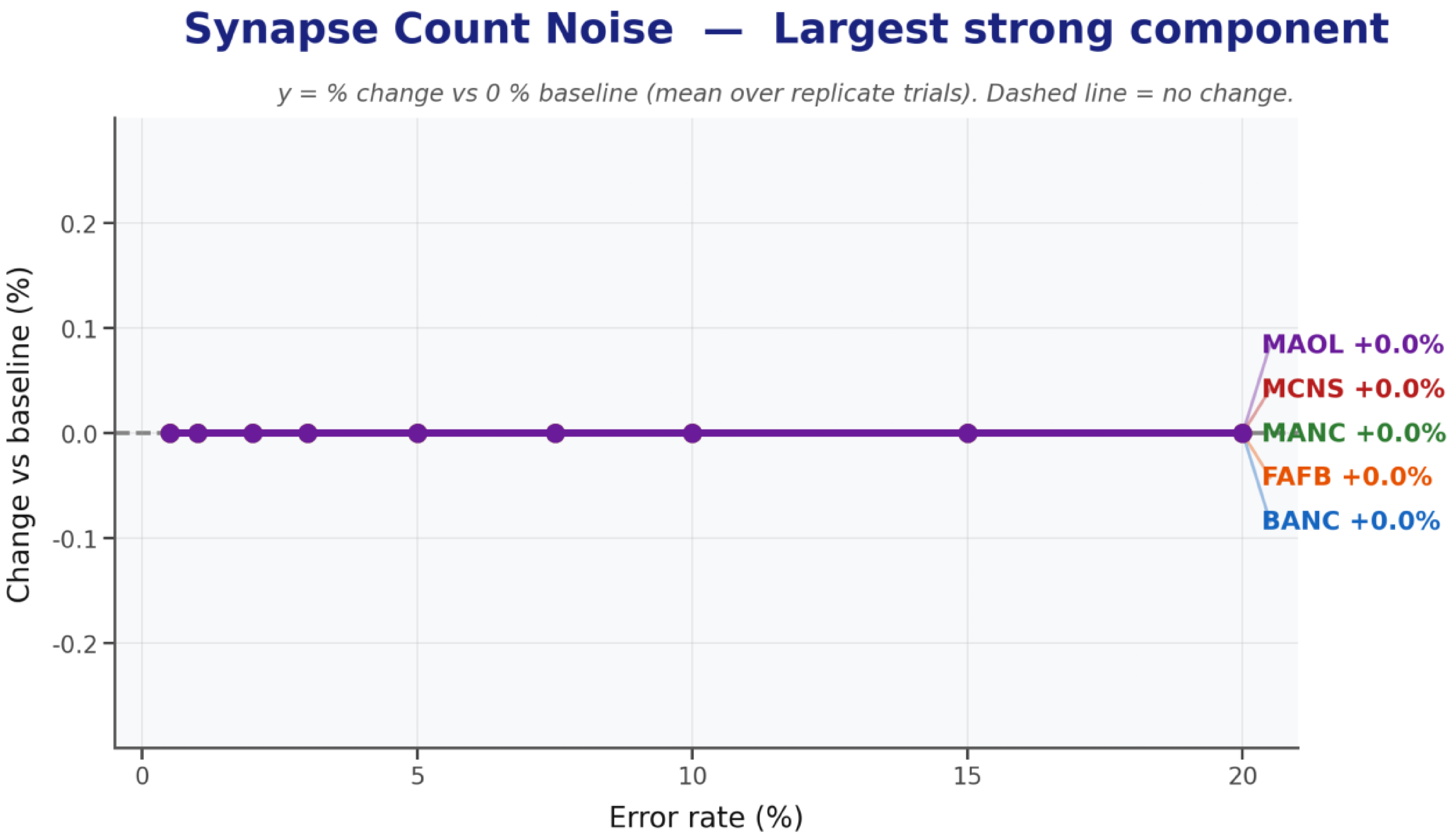


Figure 17 · Synapse count measurement — Largest strong component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

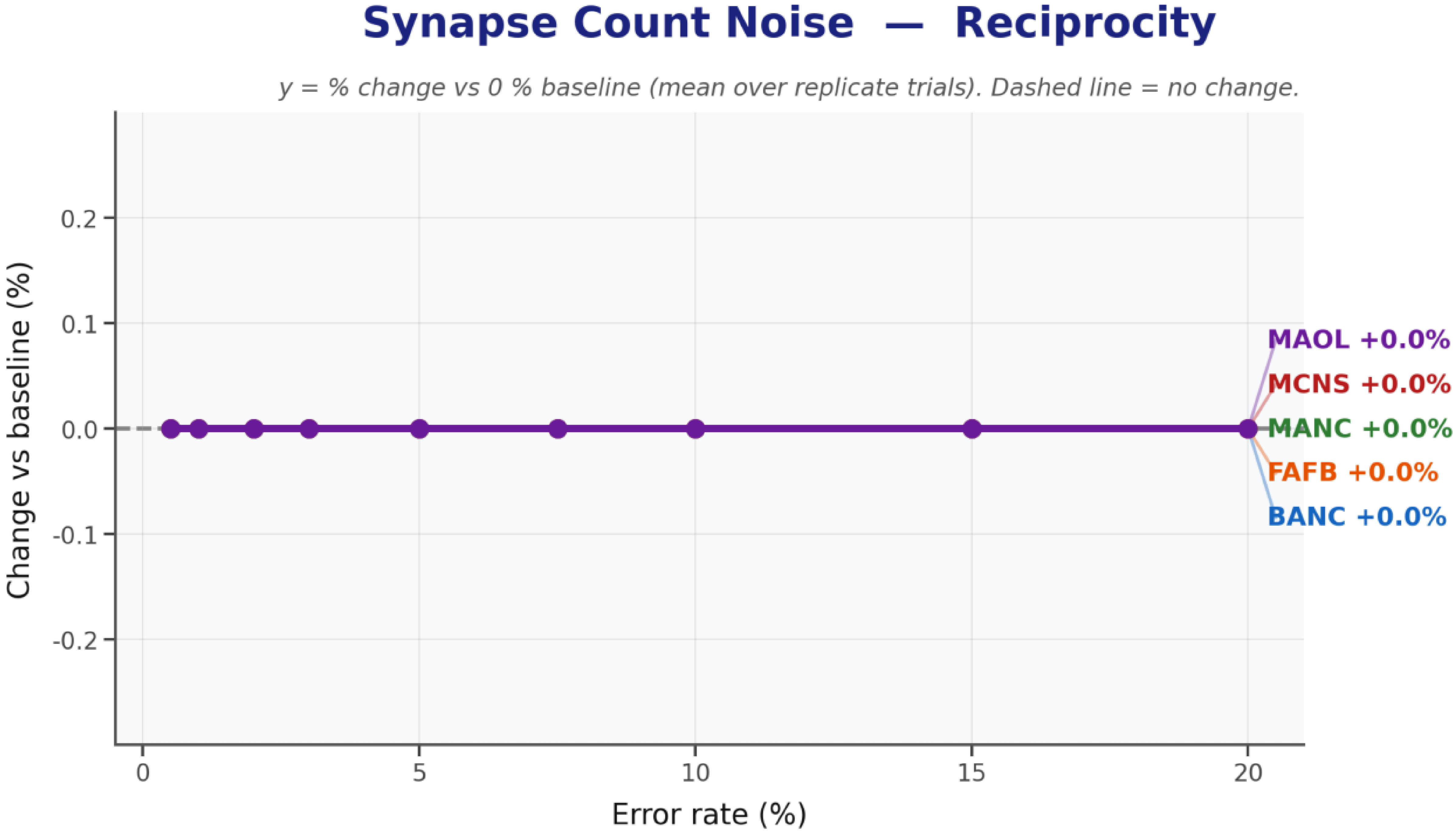


Figure 18 · Synapse count measurement — Reciprocity. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

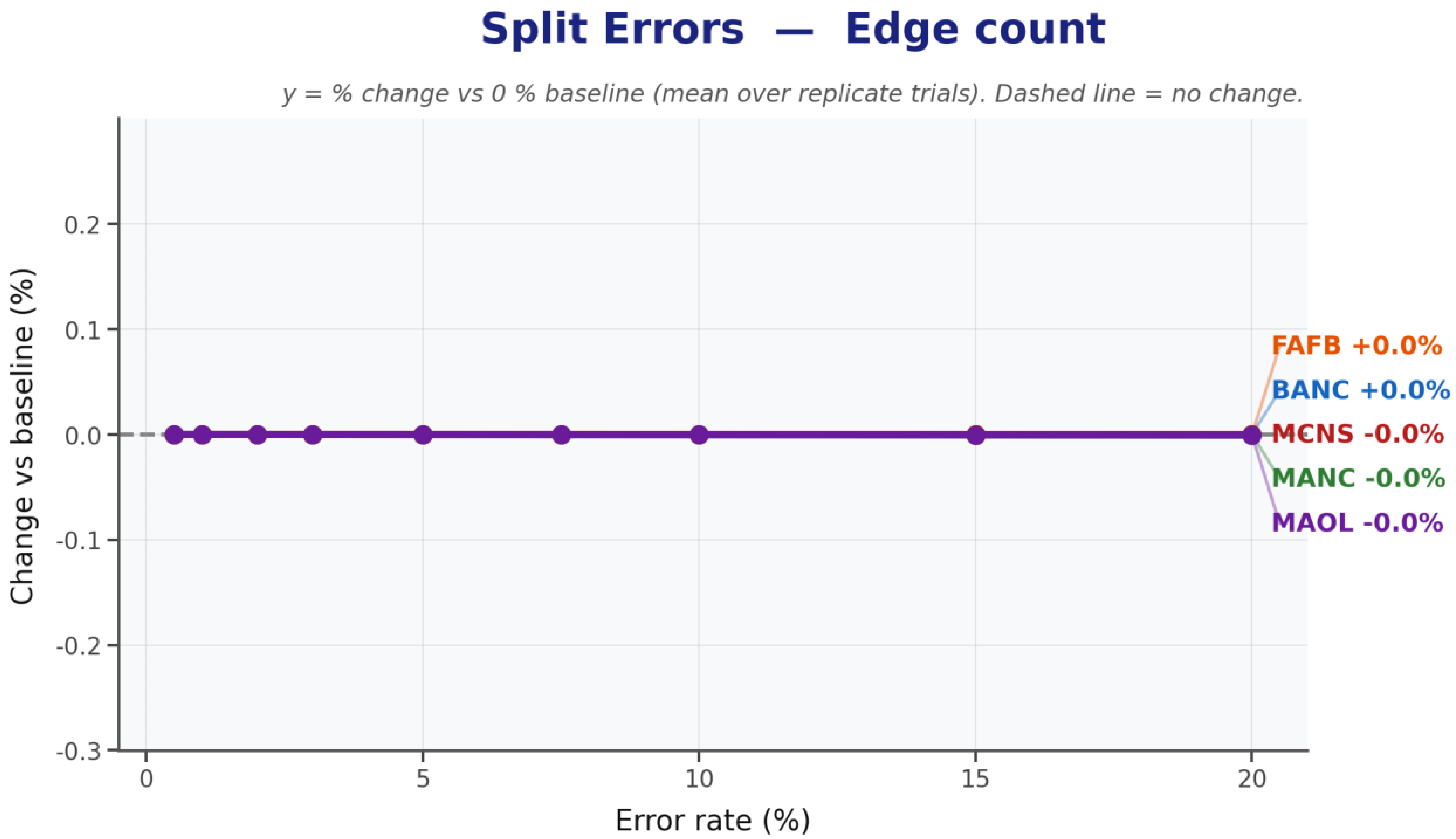


Figure 19 · Split errors — Edge count. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

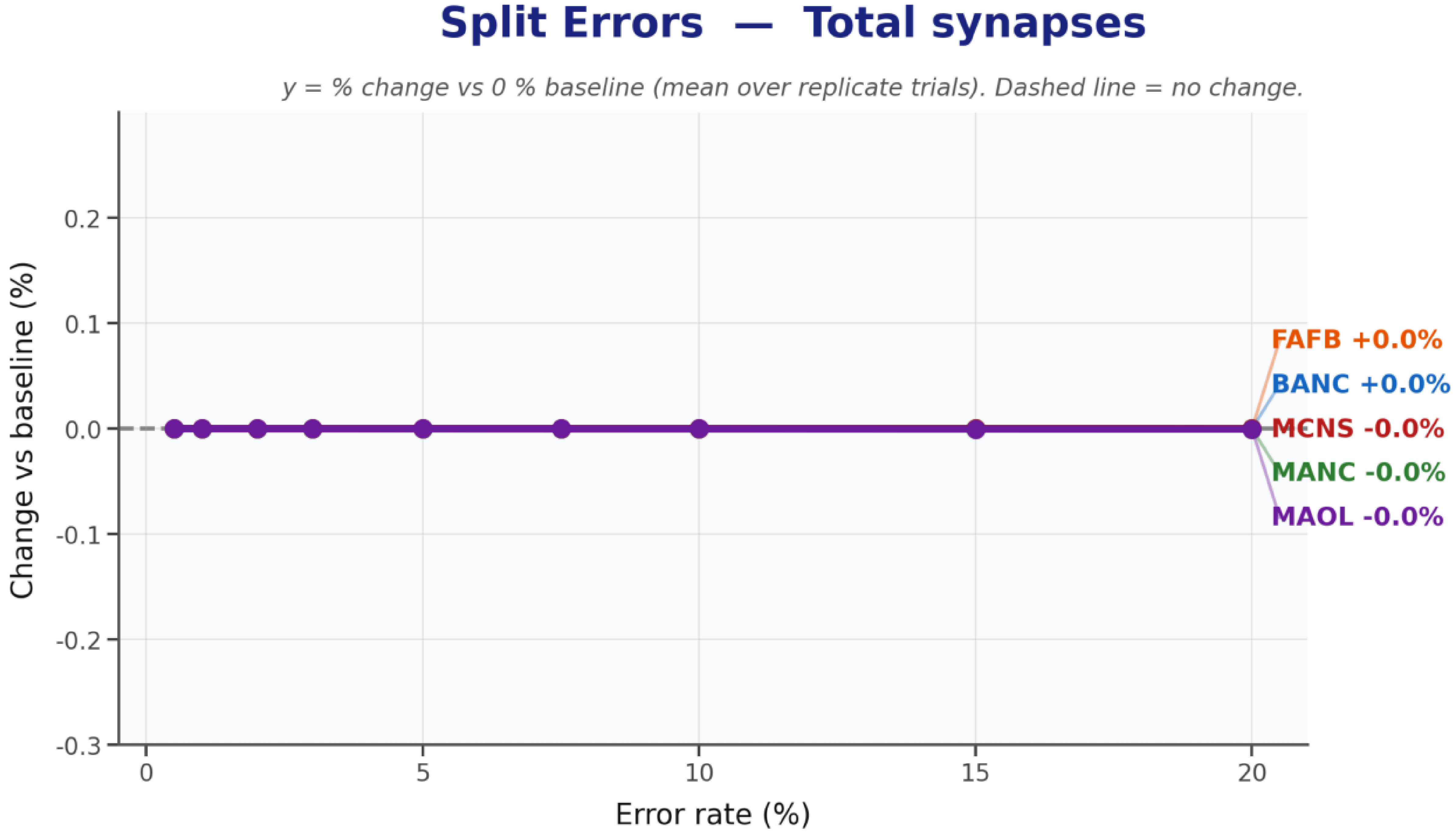


Figure 20 · Split errors — Total synapses. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

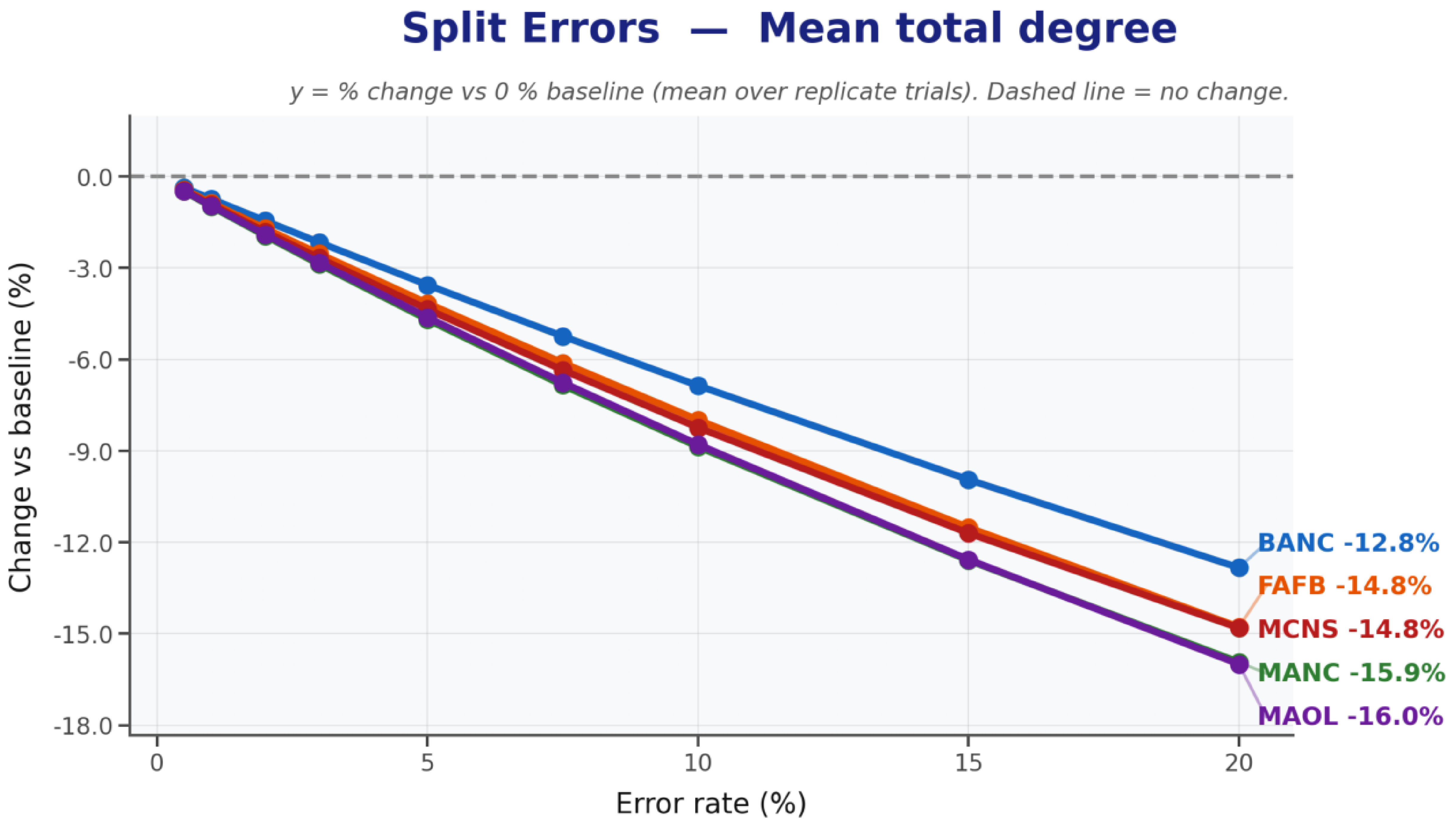


Figure 21 · Split errors — Mean total degree. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

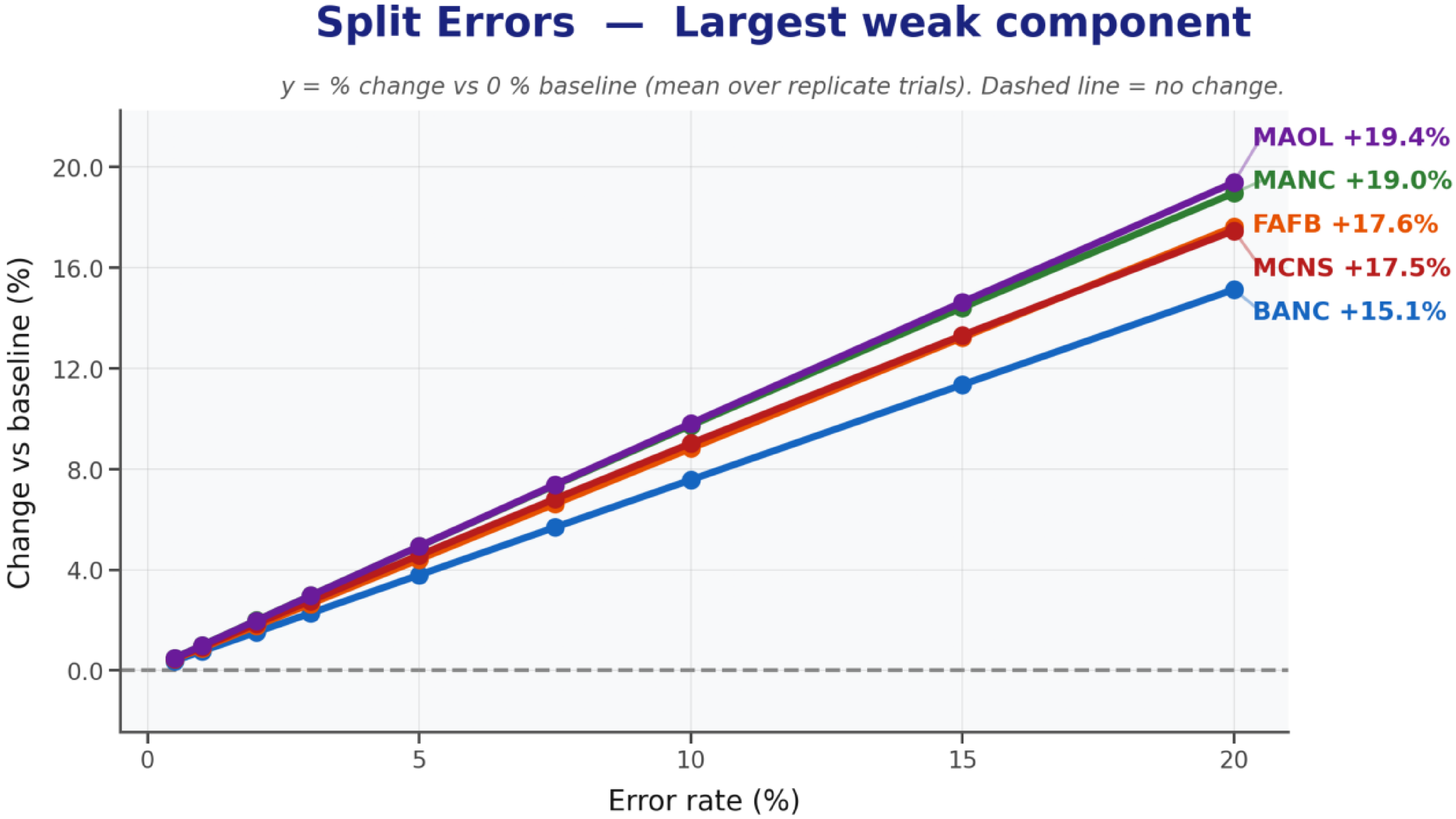


Figure 22 · Split errors — Largest weak component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

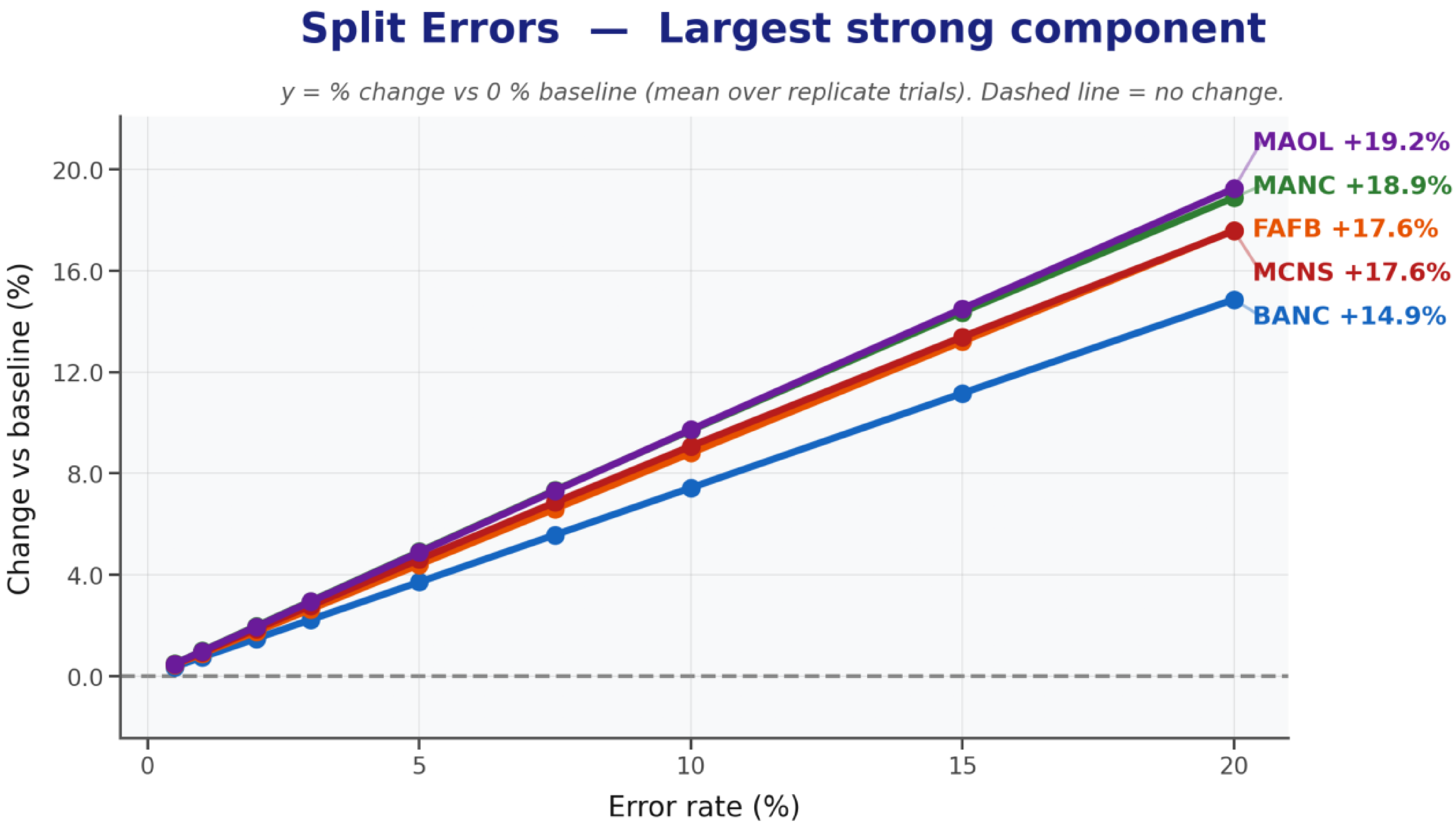


Figure 23 · Split errors — Largest strong component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

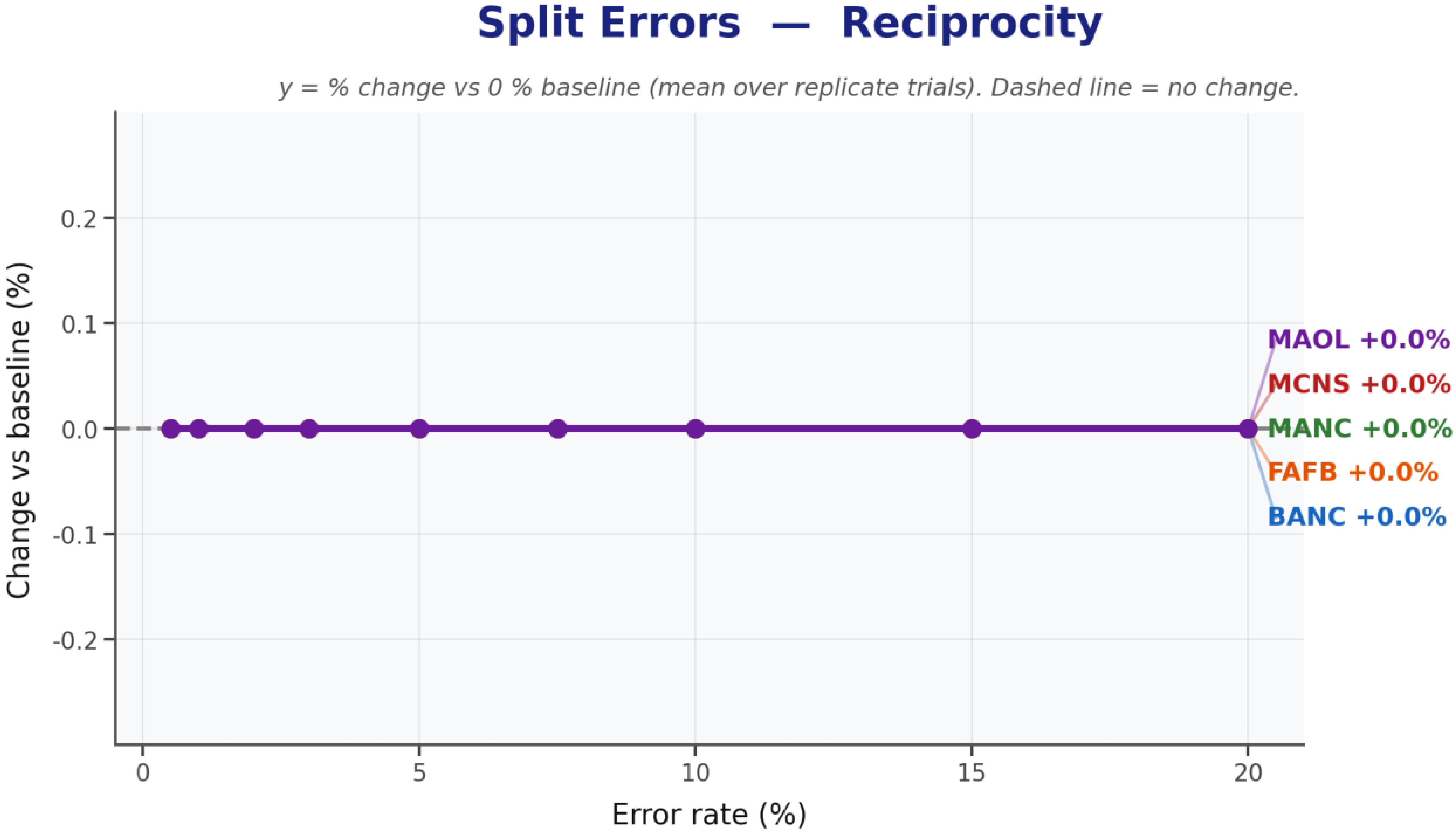


Figure 24 · Split errors — Reciprocity. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

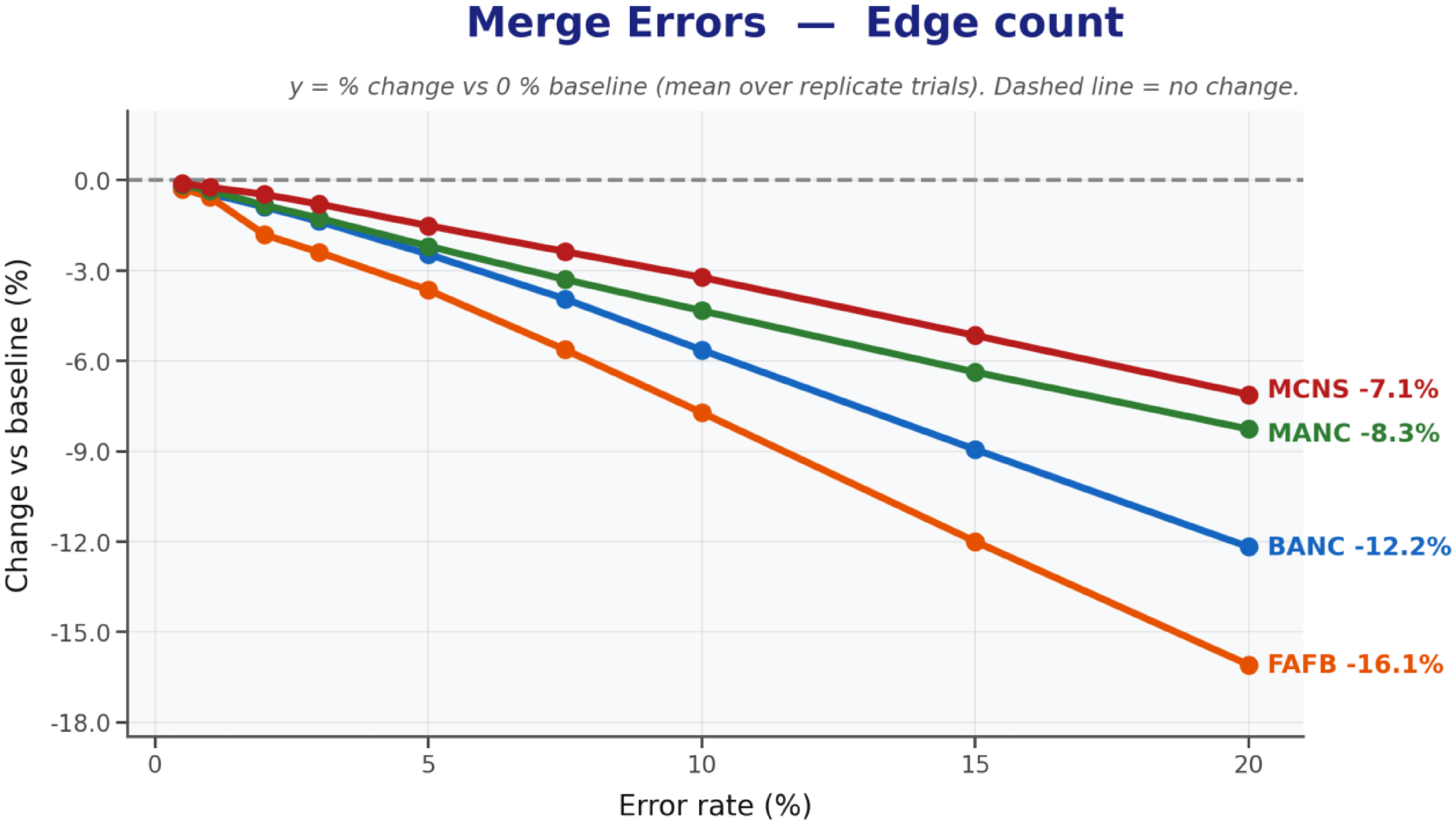


Figure 25 · Merge errors — Edge count. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

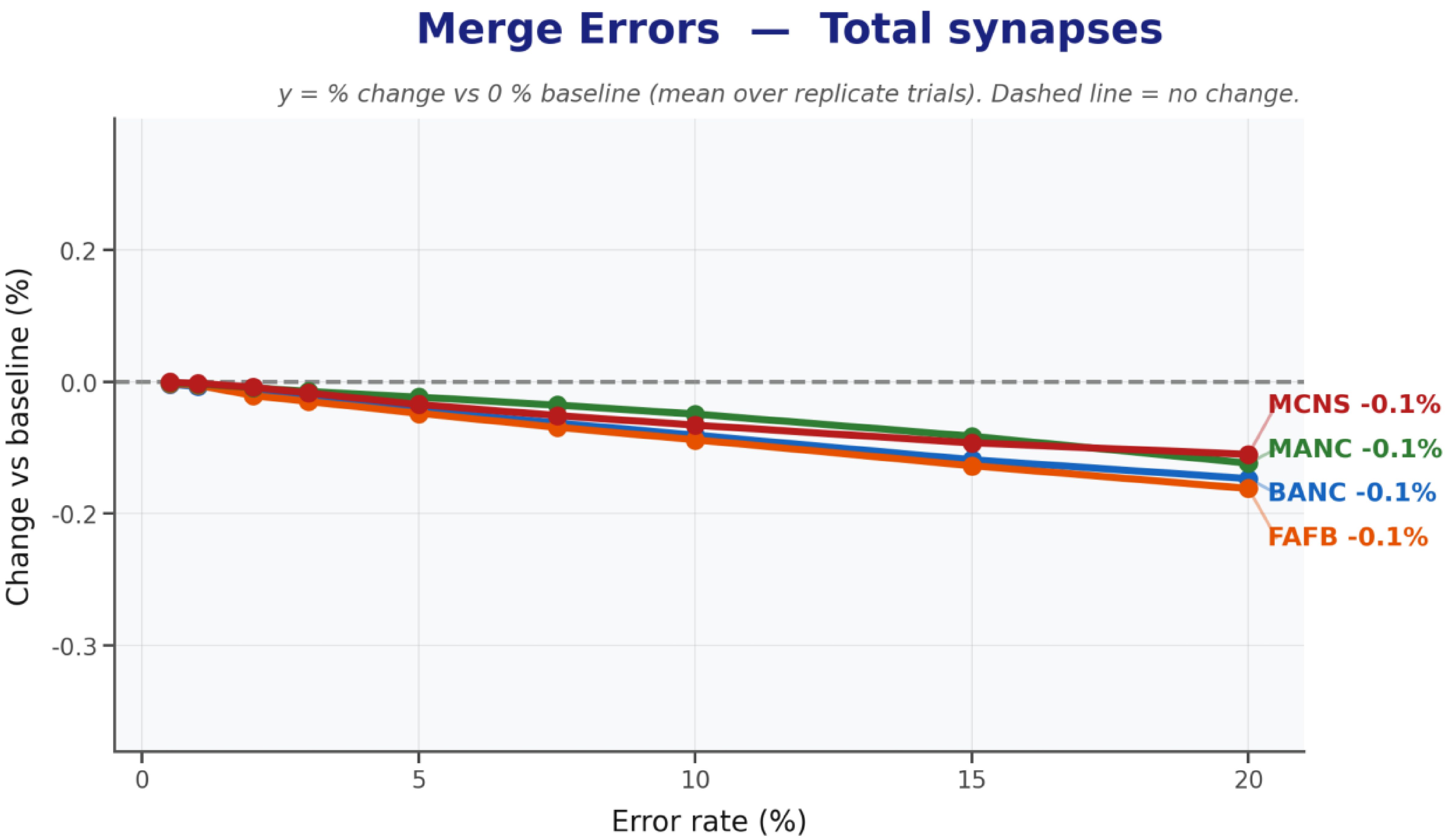


Figure 26 · Merge errors — Total synapses. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

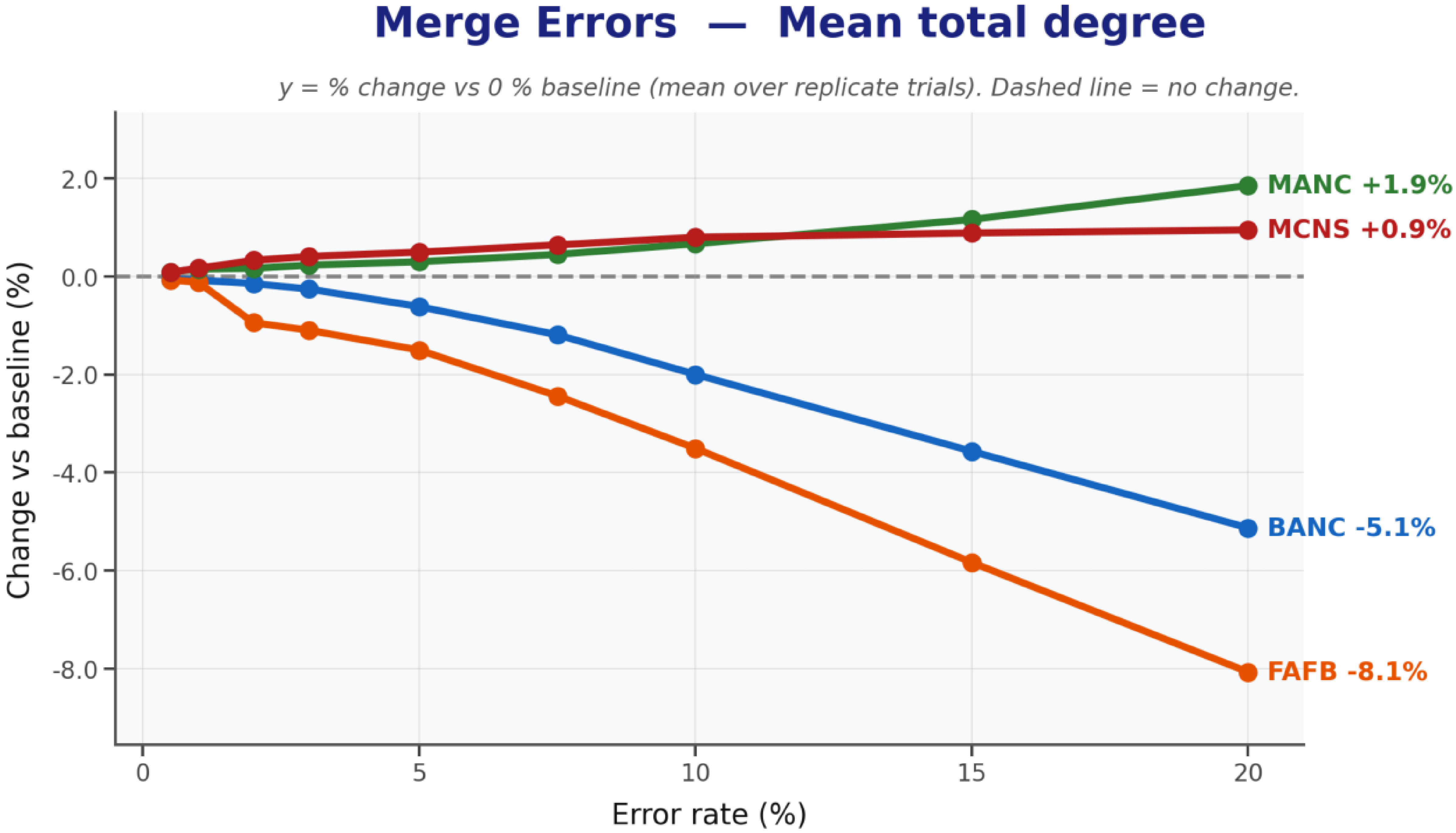


Figure 27 · Merge errors — Mean total degree. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

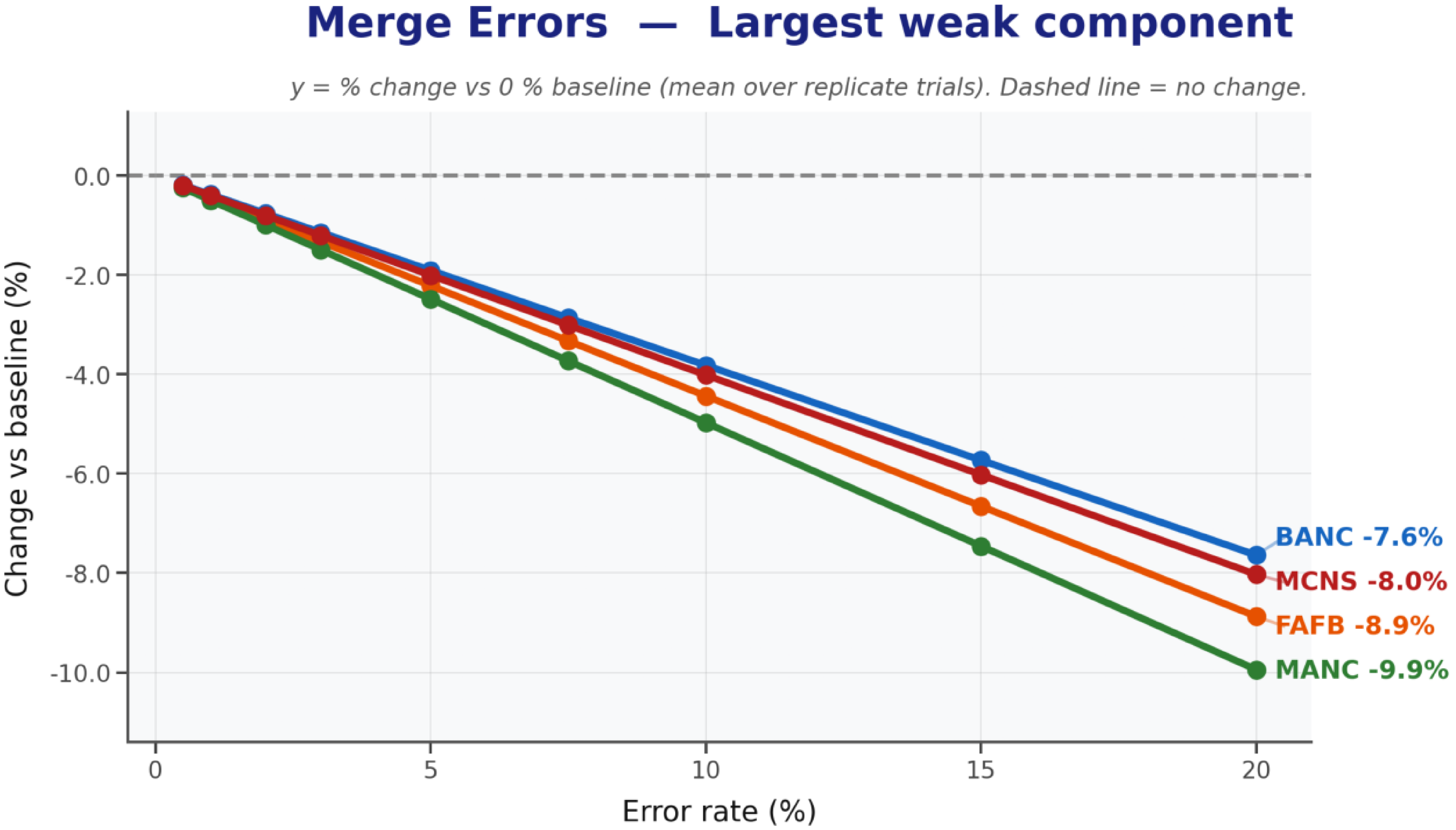


Figure 28 · Merge errors — Largest weak component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Figure 29 · Merge errors — Largest strong component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

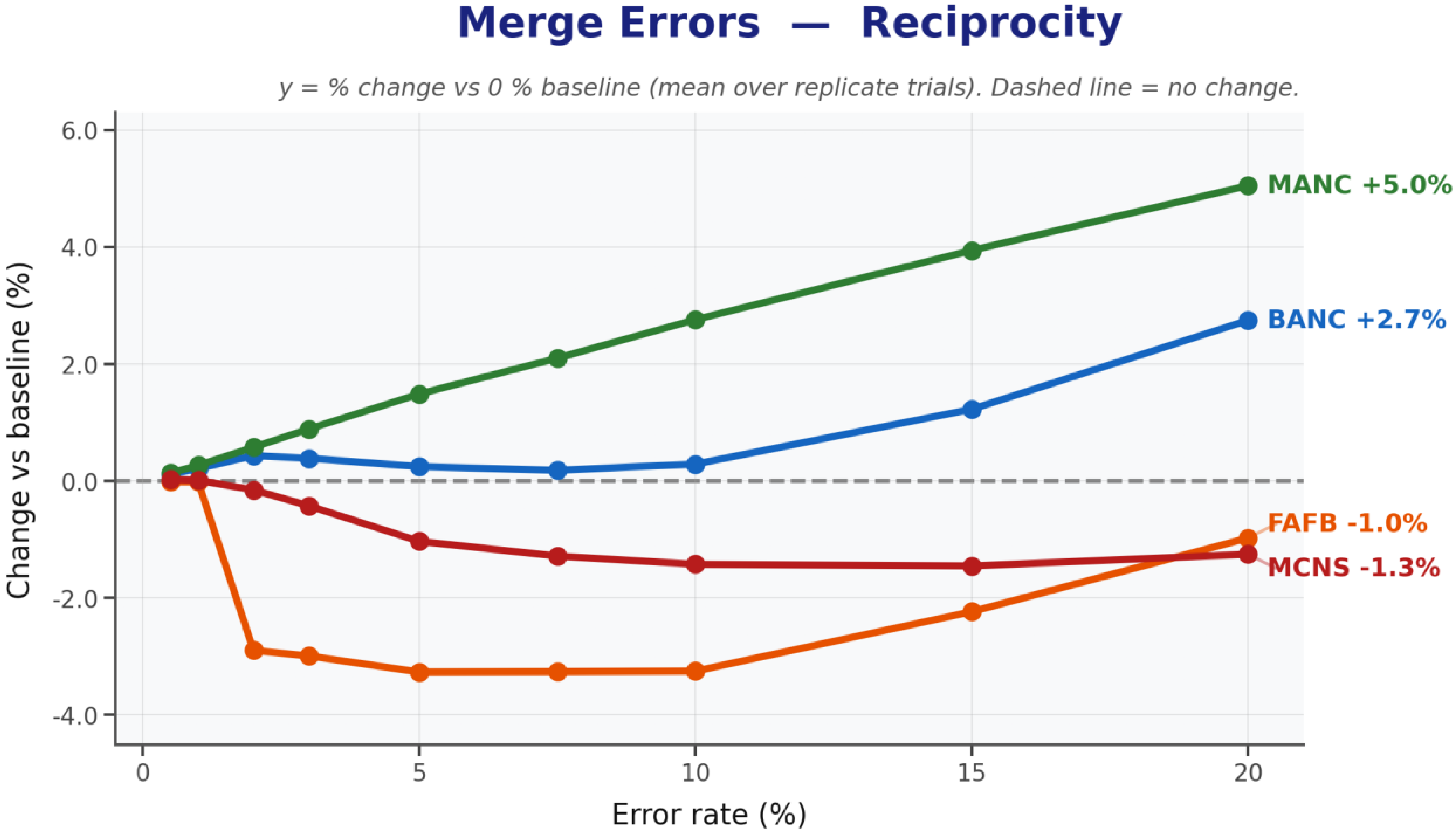


Figure 30 · Merge errors — Reciprocity. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

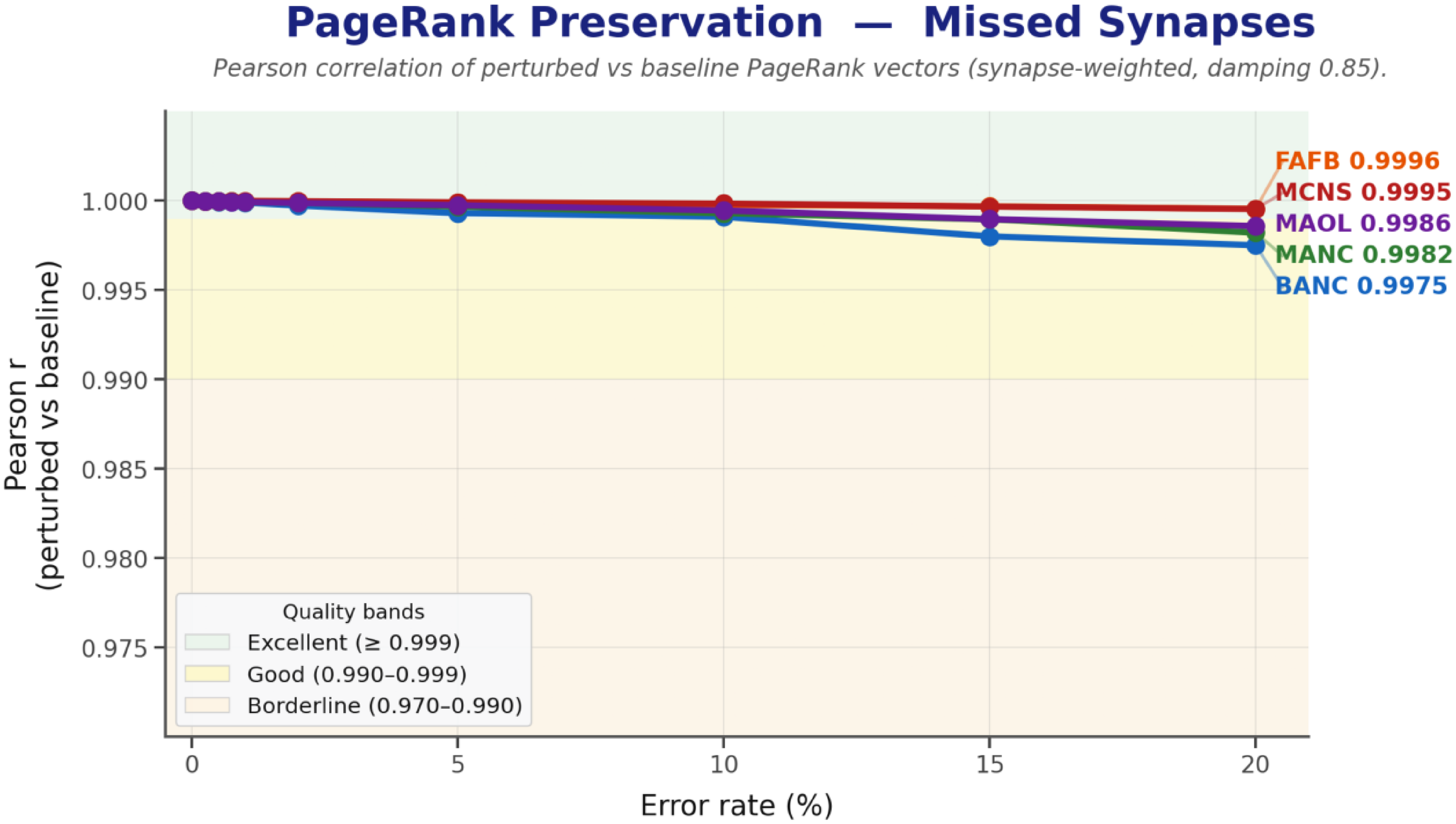


Figure 31 · PageRank preservation — Missed synapses. Coloured bands: green ≥ 0.999 (excellent), yellow 0.990–0.999 (good), orange 0.970–0.990 (borderline), red < 0.970 (degraded).

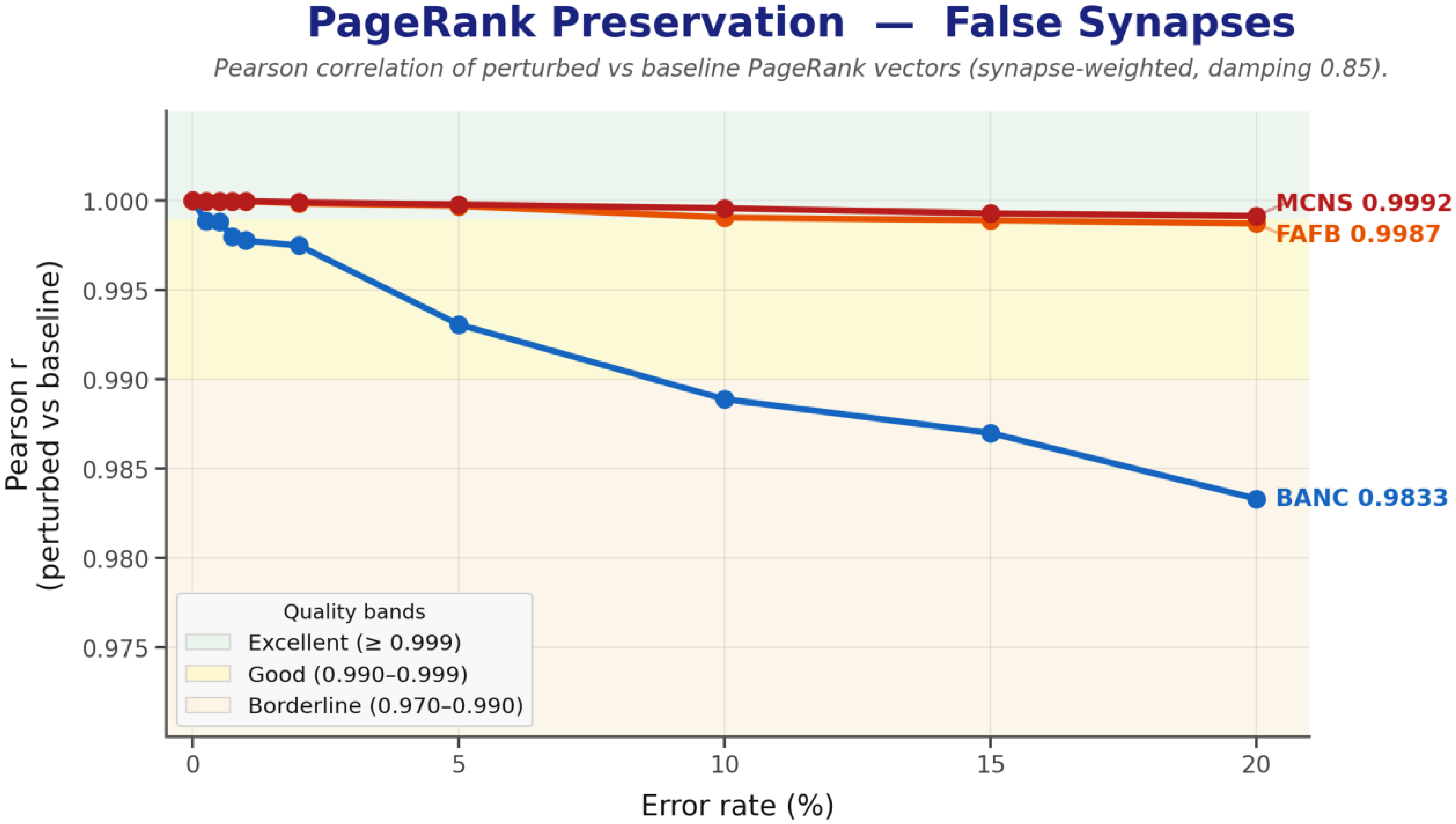
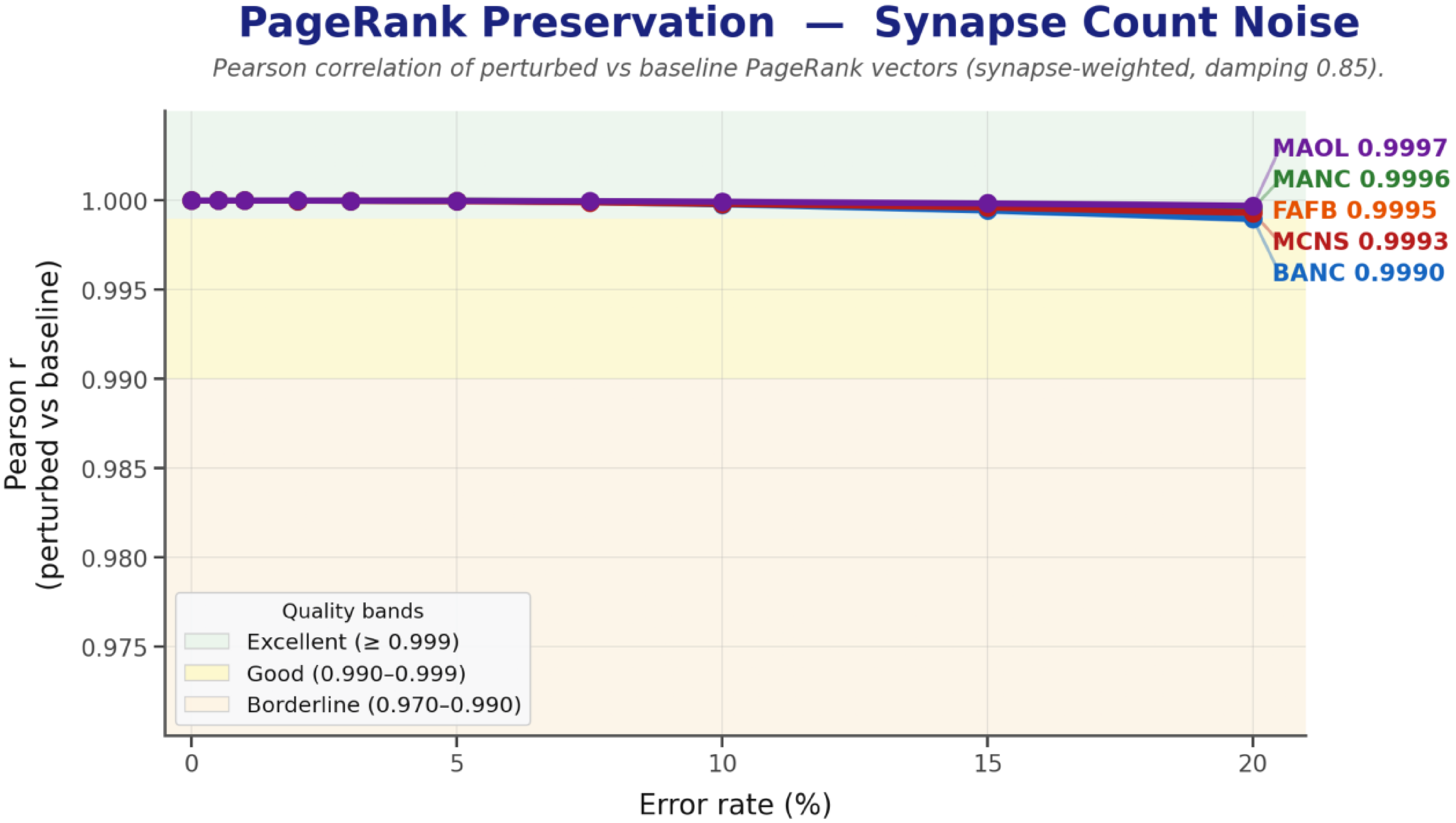


Figure 32 · PageRank preservation — False synapses. Coloured bands: green ≥ 0.999 (excellent), yellow 0.990–0.999 (good), orange 0.970–0.990 (borderline), red < 0.970 (degraded).



(degraded).

Figure 33 · PageRank preservation — Synapse count measurement. Coloured bands: green ≥ 0.999 (excellent), yellow 0.990–0.999 (good), orange 0.970–0.990 (borderline), red < 0.970

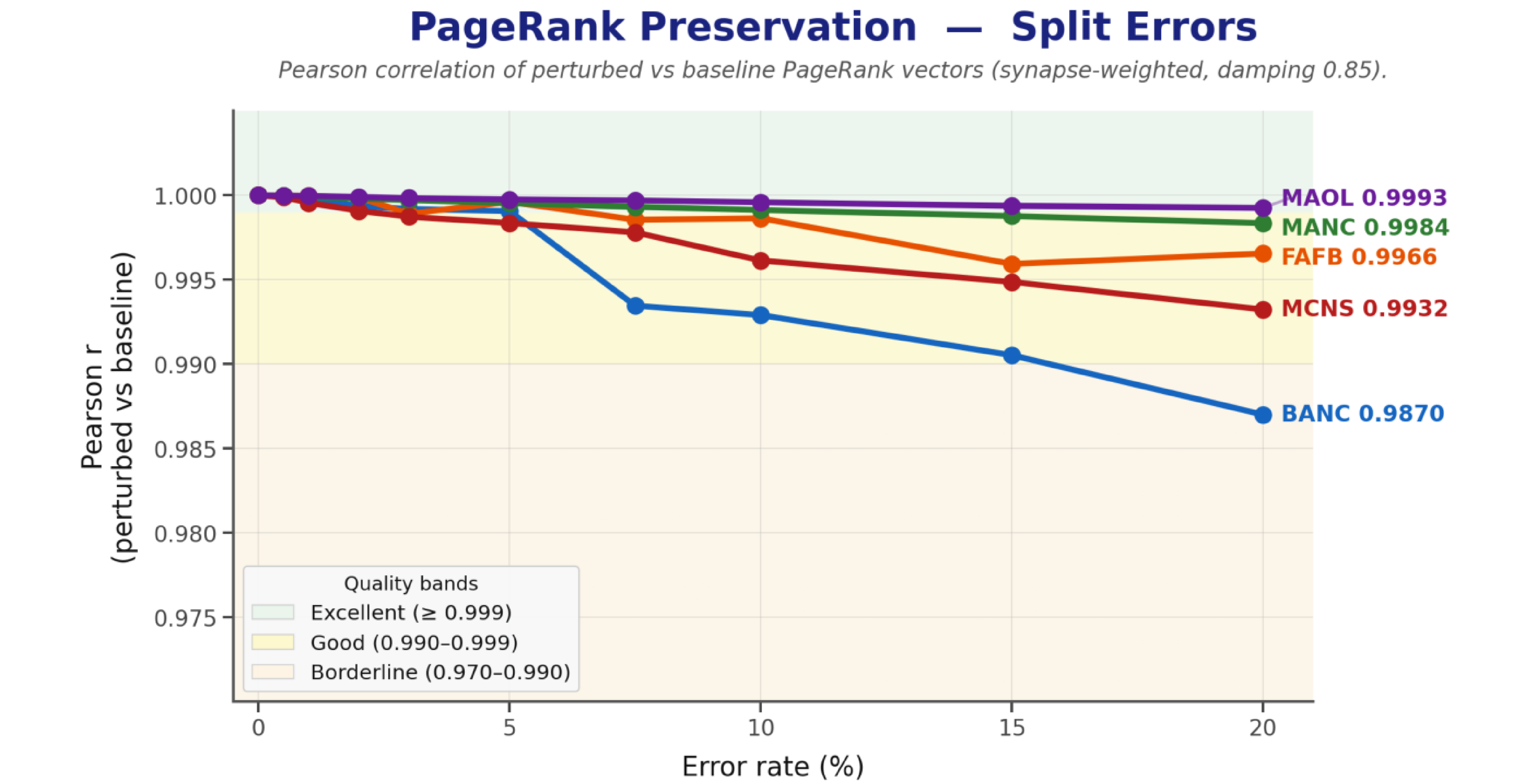


Figure 34 · PageRank preservation — Split errors. Coloured bands: green ≥ 0.999 (excellent), yellow 0.990–0.999 (good), orange 0.970–0.990 (borderline), red < 0.970 (degraded).

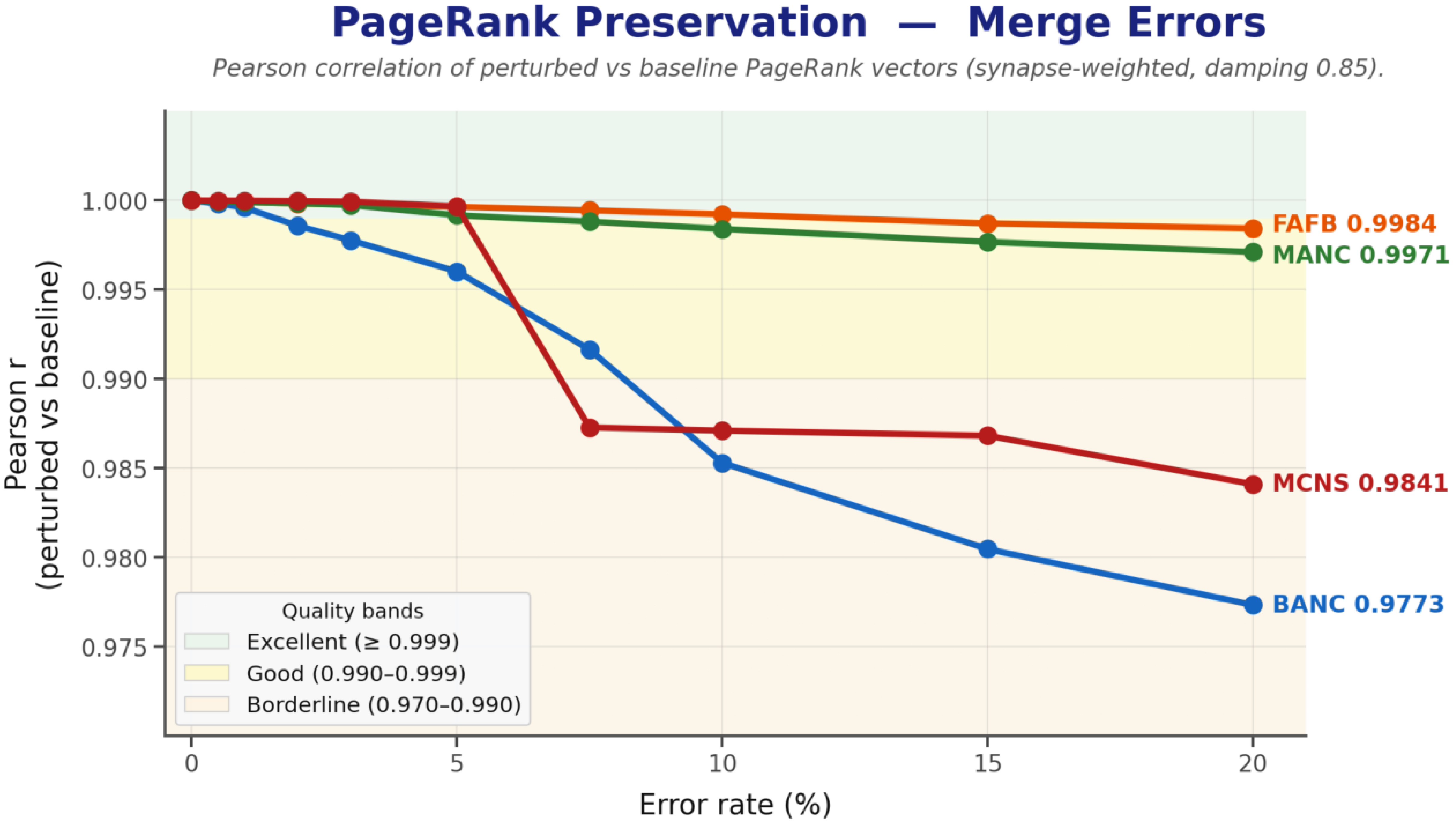


Figure 35 · PageRank preservation — Merge errors. Coloured bands: green ≥ 0.999 (excellent), yellow 0.990–0.999 (good), orange 0.970–0.990 (borderline), red < 0.970 (degraded).

§5 · Results — Worst-case Impact per Error Model

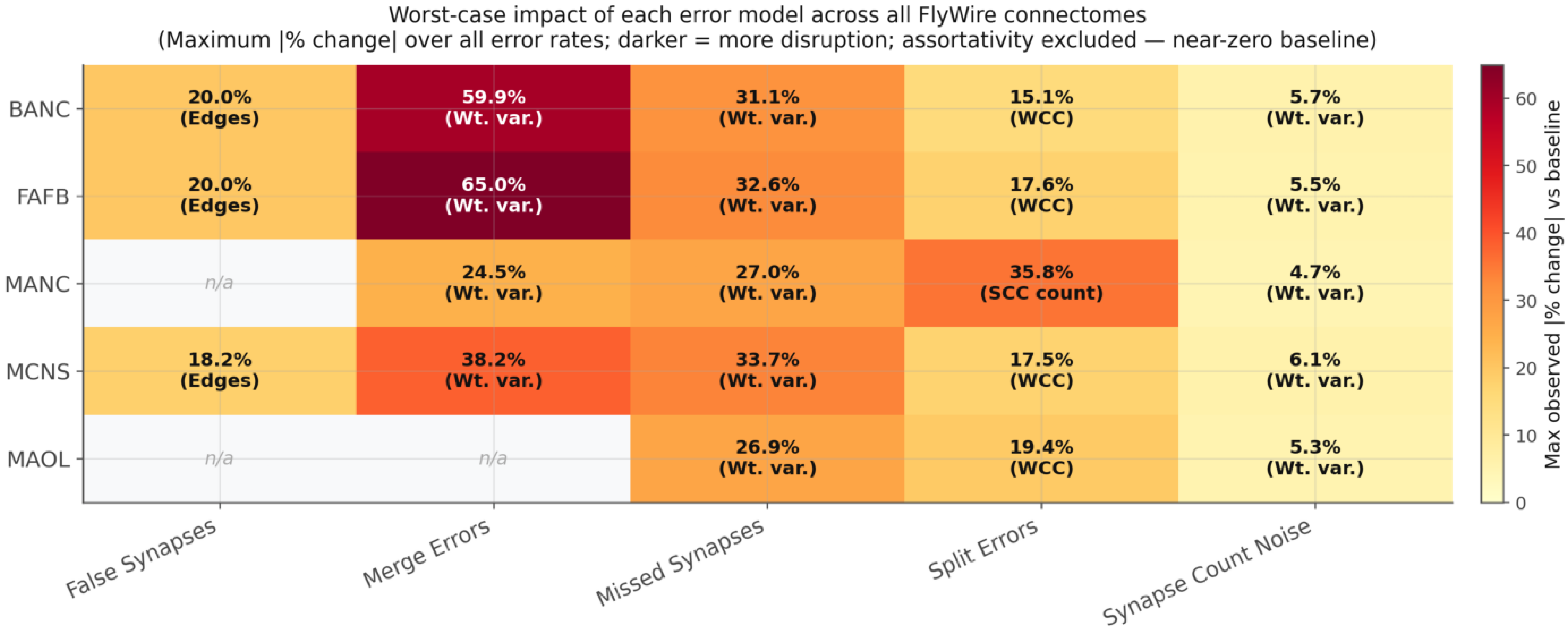


Figure 36 · Maximum observed |% change| per error model across all rates and datasets (assortativity excluded — near-zero baseline artefact). Darker cells indicate more disruption.

§5.5 · Effect Sizes at 20 % Error

Mean % change vs baseline across all datasets for each error model

Table 5 · % change at the maximum tested error rate (20 %)

Error model	Edge count	Total synapses	Weight variance	Mean degree	Largest component (WCC / SCC)	Reciprocity	Assortativity*	PageRan r
Missed synapses	-4.9%	-20.0%	-30.3%	-4.9%	WCC -0.0% SCC -0.0%	-2.9%	-3.4%	0.999
False synapses	+19.4%	+7.6%	-13.8%	+19.4%	WCC +0.0% SCC +0.6%	+5.8%	+134.9%	0.994
Synapse count measurement	+0.0%	+0.0%	+5.5%	+0.0%	WCC +0.0% SCC +0.0%	+0.0%	+0.0%	0.999
Split errors	-0.0%	-0.0%	+0.0%	-14.9%	WCC +17.7% SCC +17.6%	+0.0%	+0.3%	0.995
Merge errors	-10.9%	-0.1%	+46.9%	-2.6%	WCC -8.6% SCC -9.0%	+1.4%	-2.2%	0.989

*Assortativity has a near-zero baseline (−0.057 to +0.037; MANC is slightly positive), so its % change is inflated and should be read as an absolute movement, not a percentage. PageRank r is the mean Pearson correlation (baseline = 1.0 by construction); Spearman and top-100 overlap are reported in §5.5b. †Mean degree equals 2·edges/nodes, so its % change coincides exactly with edge count whenever the node set is unchanged; the two columns diverge only for split/merge errors, which add or remove neurons.

Spread across datasets (min … max)

Missed synapses: Edges -9.7…-0.0%; Synapses -20.0…-20.0%; Weight var. -33.7…-26.9%; Mean degree -9.7…-0.0%; WCC -0.1…+0.0%; SCC -0.1…-0.0%; Reciprocity -6.9…-0.0%; Assortativity* -9.3…+0.5%

False synapses: Edges +18.2…+20.0%; Synapses +6.3…+10.0%; Weight var. -15.0…-12.6%; Mean degree +18.2…+20.0%; WCC +0.0…+0.0%; SCC +0.3…+1.2%; Reciprocity +2.2…+8.0%; Assortativity* +16.1…+356.6%

Synapse count measurement: Edges +0.0…+0.0%; Synapses +0.0…+0.1%; Weight var. +4.7…+6.1%; Mean degree +0.0…+0.0%; WCC +0.0…+0.0%; SCC +0.0…+0.0%; Reciprocity +0.0…+0.0%; Assortativity* +0.0…+0.0%

Split errors: Edges -0.0…+0.0%; Synapses -0.0…+0.0%; Weight var. +0.0…+0.0%; Mean degree -16.0…-12.8%; WCC +15.1…+19.4%; SCC +14.9…+19.2%; Reciprocity +0.0…+0.0%; Assortativity* -10.0…+5.0%

Merge errors: Edges -16.1…-7.1%; Synapses -0.1…-0.1%; Weight var. +24.5…+65.0%; Mean degree -8.1…+1.9%; WCC -9.9…-7.6%; SCC -10.0…-8.2%; Reciprocity -1.3…+5.0%; Assortativity* -10.1…+3.2%

Missed synapses remove exactly 20 % of synapses, cutting weight variance by 30 % while deleting only the ~5 % of zero-weight edges; PageRank vectors stay highly correlated (Table 5b), so hub rankings largely survive. False synapses add ~19 % more edges, diluting weight variance by ~14 %. Synapse-count noise moves only weight variance (+5.5 %). Split errors preserve edges but spread them over more neurons. Merge errors are the most invasive (weight variance +47 %, edges −11 %). The SCC tracks the weak component except for false synapses, where new edges grow the strong core (+0.6 %).

§5.5b · PageRank Preservation at 20 % Error

Pearson, Spearman and top-100 overlap — all three measures present in the exports

Table 5b · PageRank preservation at 20 % error — mean across datasets (min–max across datasets)

Error model	Pearson r	Spearman ρ	Top-100 overlap
Missed synapses	0.999 (0.998–1.000)	0.998 (0.996–0.999)	0.980 (0.970–0.986)
False synapses	0.994 (0.983–0.999)	0.971 (0.962–0.980)	0.951 (0.892–0.982)
Synapse count measurement	0.999 (0.999–1.000)	0.999 (0.998–1.000)	0.976 (0.970–0.986)
Split errors	0.995 (0.987–0.999)	0.976 (0.969–0.986)	0.956 (0.942–0.978)
Merge errors	0.989 (0.977–0.998)	0.989 (0.986–0.991)	0.924 (0.860–0.960)

Pearson correlation measures vector similarity, not rank preservation. Spearman correlation measures rank-order agreement, and top-100 overlap measures how many of the 100 highest-PageRank neurons in the baseline remain in the top 100 after perturbation. Across all models Pearson r ≥ 0.977 and Spearman ≥ 0.96 at 20 % error, so PageRank vectors remain highly correlated and rank order is largely, but not entirely, retained. The largest displacements occur on BANC: under merge errors the top-100 overlap falls to 0.86 and under false synapses to 0.89, i.e. roughly one in seven of the top hubs is displaced from the baseline top 100.

§5.6 · Per-Dataset Breakdown at Peak Error Rate

How variation across connectomes reinforces the headline conclusions

Table 6 · Edge count and total-synapse change per dataset at 20 % error, with PageRank Pearson r (FAFB and MANC merge rows average 2 and 1 trials per rate; all other rows average 5)

Error model	Dataset	Edge count Δ%	Total synapses Δ%	PageRank r
Missed synapses	BANC	-3.2%	-20.0%	0.998
Missed synapses	FAFB	-2.5%	-20.0%	1.000
Missed synapses	MANC	-9.7%	-20.0%	0.998
Missed synapses	MCNS	-0.0%	-20.0%	1.000
Missed synapses	MAOL	-8.9%	-20.0%	0.999
False synapses	BANC	+20.0%	+10.0%	0.983
False synapses	FAFB	+20.0%	+6.6%	0.999
False synapses	MCNS	+18.2%	+6.3%	0.999
Synapse count measurement	BANC	+0.0%	+0.0%	0.999
Synapse count measurement	FAFB	+0.0%	+0.0%	0.999
Synapse count measurement	MANC	+0.0%	+0.1%	1.000
Synapse count measurement	MCNS	+0.0%	+0.0%	0.999
Synapse count measurement	MAOL	+0.0%	+0.1%	1.000
Split errors	BANC	+0.0%	+0.0%	0.987
Split errors	FAFB	+0.0%	+0.0%	0.997
Split errors	MANC	-0.0%	-0.0%	0.998
Split errors	MCNS	-0.0%	-0.0%	0.993
Split errors	MAOL	-0.0%	-0.0%	0.999
Merge errors	BANC	-12.2%	-0.1%	0.977
Merge errors	FAFB	-16.1%	-0.1%	0.998
Merge errors	MANC	-8.3%	-0.1%	0.997
Merge errors	MCNS	-7.1%	-0.1%	0.984

This table shows that the cross-dataset means in Table 5 do not hide large disagreements between connectomes. The −4.9 % average edge loss under missed synapses spans −0.0 % (MCNS) to −9.7 % (MANC): datasets whose edges carry fewer synapses lose edges faster, because an edge survives only if at least one of its synapses survives (Table 6b). Merge errors reduce edge count in every dataset at 20 % (FAFB −16.1 % to MCNS −7.1 %); no dataset shows an increase.

§5.6b · Why Edge Loss Differs Across Connectomes

Edge-weight distribution vs edge loss at 20 % synapse loss (missed-synapse model)

Table 6b · Baseline edge-weight distribution vs edge loss under the missed-synapse model at 20 %

Dataset	Mean synapses/edge	Median synapses/edge	Edge count Δ% at 20 % synapse loss
BANC	5.9	4	-3.2%
FAFB	9.5	6	-2.5%
MANC	5.0	2	-9.7%
MCNS	14.4	9	-0.0%
MAOL	3.9	2	-8.9%

An edge survives a missed-synapse run only if at least one of its synapses survives, so at a fixed synapse-loss rate the fraction of edges deleted is largest for datasets whose edges carry the fewest synapses. The median synapses/edge tracks the observed edge loss monotonically across the five connectomes: MANC and MAOL (median 2) lose −9.7 % and −8.9 % of edges, BANC (median 4) −3.2 %, FAFB (median 6) −2.5 %, and MCNS (median 9) −0.0 %. This is a descriptive relationship across five connectomes (n = 5); it is consistent with the binomial-deletion mechanism but is not claimed as a general law.

§6 · Conclusions

Practical implications for segmentation benchmarking

- 1. Disproportionate Impact of Connection Loss: Missed synapses remove synapses proportionally with the error rate and erode weight variance by 30 % at 20 %, while false synapses primarily inflate edge counts. Algorithmic benchmarking should prioritize missed-synapse detection.
- 2. Resilience to Measurement Noise: Synapse-count uncertainty perturbs weight statistics but leaves the overarching topology intact. Graph analyses unweighted by synapse count remain unaffected.
- 3. Structural Shifts via Mis-segmentation: Split and merge errors alter the neuron count rather than just edge topology. Analyses treating neurons as fundamental units (e.g., community detection, PageRank) are highly sensitive to these errors.
- 4. PageRank Robustness: Despite local topological degradation, PageRank vectors remain highly correlated at a 20 % error rate (Pearson r ≥ 0.977 and Spearman ≥ 0.96 for every model and dataset). Most, but not all, high-centrality neurons retain their rank: top-100 hub overlap stays ≥ 0.86, with the largest displacements under false and merge errors on BANC (Table 5b). Local measures (degree, component size) act as earlier indicators of structural decay.
- 5. Consistency Across the Evaluated Connectomes: The five evaluated Drosophila connectomes exhibit qualitatively similar responses to identical error models. Broader generalization is not claimed: all datasets come from one species, and coverage is partial (Table 2).

Caveats

FAFB and MANC merge-error results rest on 2 and 1 trials per rate (partial runs in the source archives); MANC and MAOL have no false-synapse run. All other cells use 5 replicate trials, and trial-to-trial SD of the reported % changes stays below 0.18 pp in every 5-trial cell for the structural headline metrics (weight variance reaches 1.3 pp and assortativity 14.9 pp). The %-change framing inflates metrics with near-zero baselines (assortativity) — always check the raw direction and magnitude. The five error models are computational simulations: their error distributions are not empirically validated against real reconstruction-error ground truth (no such data is part of this repository), so the results describe how the modelled error classes perturb graph structure, not measured segmentation errors. 1,030 trials in total; every number in this report was recomputed from the per-trial CSVs.

§7 · Reproducibility

How to regenerate this report from source data

The whole report — CSVs, figures and this PDF — is produced by three scripts in the analysis/ folder. Run them in order from the repository root:

```
$ python analysis/01_load_and_verify.py
```

```
$ python analysis/01_load_and_verify.py
loaded 6180 analysis-rows, 5 datasets, 5 error models, 10 rates, max 5 trials per cell

[1] COMPLETENESS
  trials per cell: min=1, max=5, cells=220
  PARTIAL FAFB/merge_errors: only 2 trials/rate
  PARTIAL MANC/merge_errors: only 1 trial/rate
  analyses per trial: min=6, max=6 (expect 6)
  non-SUCCESS rows: 0

[2] BASELINE INVARIANCE (0% error)
  BANC: baseline max rel diff across error models = 0.00e+00
  FAFB: baseline max rel diff across error models = 0.00e+00
  MANC: baseline max rel diff across error models = 0.00e+00
  MCNS: baseline max rel diff across error models = 0.00e+00
  MAOL: baseline max rel diff across error models = 0.00e+00

[3] INTERNAL CONSISTENCY
  in-degree mean vs edges/nodes: max rel err = 2.206e-16
  density vs edges/(n*(n-1)):    max rel err = 6.978e-13
```

Appendix · Statistical Notes

Known artefacts and how they are handled in this report

- Near-zero-baseline % change. Relative change is undefined in spirit when the baseline is near zero. Assortativity (baseline −0.057 to +0.037; MANC is slightly positive) shows % swings of +135 % for a small absolute movement; we report the raw sign and magnitude and exclude the metric from the heatmap.
- Reciprocity baselines (0.13–0.29) are safe for % change; the biggest move is merge errors (+1.4 % mean).
- Mean-over-datasets. Table 5 averages % changes across the datasets that were run. Table 6 gives every dataset individually, and the spread (min…max) is printed under Table 5.
- Partial runs. FAFB merge (2 trials/rate), MANC merge (1 trial/rate); MANC/MAOL false synapses and MAOL merge not run. Figures simply omit missing lines.
- Deterministic vs stochastic. False-synapse edge additions are deterministic ($k = \mathrm{round}(r \cdot E)$ every trial), so their trial-to-trial spread is zero — an identity, not a bug (verified in §4).